Triangle Hypersurfaces and a Sharp Identifiability Boundary for Level-2 K3P Networks

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Abstract

We classify regular full-dimensional stochastic containment among binary standard semi-directed strongly tree-child level-2 phylogenetic networks under the Kimura three-parameter (K3P) model. On the principal positive Fourier domain, a directed containment germ exists if and only if the labelled reduced trees of blobs agree and corresponding complete factors are either labelled-isomorphic or ordinarily triangle-redirected, with coherent boundary transports. The same condition is equivalent to a common full-dimensional regular germ, and it remains exact in strict continuous time. Thus no proper one-sided containment occurs in the strong class, and the semi-directed topology is generically identifiable and, outside the same proper exceptional set, exactly reconstructible modulo ordinary triangle redirection.

K3P has a distinct local geometry. The three ordinary triangle orientations have generic normalized rank 14, share the same irreducible eight-term quartic hypersurface $H_{14} \subset \mathbb{A}^{15}$, and meet in a common strict continuous-time smooth rank-14 germ; they do not fill an ambient-open three-leaf germ. Necessity combines three-sector bridge fibres, physical marginal submersions, a noncircular bridge-split transfer theorem, and an exact finite classification. The cut argument deliberately does not assert a universal pointwise K3P rank criterion: true cuts have rank at most four pointwise, noncuts are generically detected, and equality of cut sets is proved under source-relative containment in the strong class by 204 pointwise one-active obstructions. The remaining bounded residue consists of fourteen four-port orbits—nine polynomially separated and five directed-rank separated—plus two separately separated sink swaps.

The strong hypothesis is sharp. For every $n \geq 3$, two weakly but not strongly tree-child networks have strict continuous-time K3P images sharing a common full-dimensional regular germ of dimension $6n - 3$. The base rank-15 common point is certified by exact rational interval arithmetic and a Krawczyk inclusion. Exact certificates, separately implemented replays, and fail-closed mutation tests accompany every finite and interval claim.

Keywords. phylogenetic networks; Kimura three-parameter model; generic identifiability; directed containment; level-2 networks; strongly tree-child networks; algebraic statistics; computer-assisted proof.

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1 Introduction

Phylogenetic networks represent evolutionary histories in which lineages can merge as well as split. Their statistical models are structured mixtures: at each reticulation one parent is selected, and conditional on all such choices the site evolves on a displayed tree. Identifiability is therefore an information-limit question. Even with an exact site-pattern distribution—the infinite-data idealization—can two genuinely different network topologies describe the same full-dimensional family of observations?

The distinction between a shared point and a shared regular germ is central. A boundary degeneration, a singular intersection, or equality after complexification does not by itself create a robust stochastic ambiguity. Conversely, comparing generic dimensions alone does not exclude a directed containment when a lower-dimensional source can enter an exceptional locus of a larger target. We accordingly study a source-relative analytic containment with a physical target section. This relation records an open family of exact stochastic coincidences and retains its direction; symmetry is a conclusion of the classification, not part of the definition.

Group-based Fourier coordinates make the Kimura family especially suitable for exact analysis [9, 22]. In the K3P model the three nonzero characters C, G, T have independent edge multipliers. This is both the model's biological flexibility and its main algebraic difficulty: bridge fibres have three independent incidence scales, marginals carry three product coordinates, and identities that were one-sector under JC or two-sector under K2P need not survive. The K3P problem is thus not a formal substitution into either companion proof.

For level one, algebraic invariants and Jacobian methods separate broad network classes and isolate the familiar ambiguity of redirecting an ordinary triangle [11, 13, 12, 10, 5, 4]. Gross, Krone, and Martin computationally record the K3P three-sunlet's one-dimensional deficiency. Cox, Gross, and Martin prove a dimension theorem for odd-order group-based three-sunlets and computationally study the even-order cases. For JC, Currie et al. give complete semialgebraic descriptions of the three oriented triangle models and show that their pairwise intersections and set differences are full-dimensional [7]; that ambient JC geometry is the closest analogue of, but is distinct from, the relative K3P germ on H_{14} proved here. Complementary multigraded implicitization methods compute low-degree invariant spaces for K3P four- and five-leaf sunlet parameterizations by graded linear algebra, including a four-leaf cubic calculation for which a reported degree-limited Gröbner computation had not produced all generators after 100 days [6]. Gross et al. already observe that the ideals of the three K3P triangle orientations agree [12]. The new exact local refinements here are a certified normalized rank-14 theorem, an explicit irreducible eight-term defining quartic for that common closure, and a common strict smooth physical germ; the global contribution is to place that germ inside a complete level-two containment classification. After translating Brits et al.'s exhaustive cleanup convention to our one-step root suppression and restricting to standard-admissible presentations, the resulting triangle quotient agrees with their pointwise full-identifiability theorem for level-one semi-directed networks modulo triangle redirection under JC, K2P, and K3P.

At level two, invariant calculations for simple and semisimple networks and quartet-based combinatorial reconstruction give complementary partial results [2, 14]. Englander et al. establish generic JC identifiability for binary, triangle-free, strongly tree-child level-2 semi-directed networks and give pointwise displayed-quartet tools under JC and K2P [8]. Removing triangle-freeness and classifying the resulting triangle ambiguity are therefore part of the present advance. The practical invariant literature also includes a four-taxon JC/K2P method tested on simulated and biological sequence data [20]; it addresses finite-data inference for four-cycle networks rather than the exact local-to-global containment problem here. Complementarily, Allman et al. obtain identifiability for large, explicitly restricted subclasses of binary galled tree-child networks, including nonplanar arbitrary-level examples, from quartet concordance factors under gene-tree models and additional genericity, sampling, and small-blob hypotheses [1]; their data, model, and topology hypotheses are different and do not subsume displayed-tree K3P site-pattern containment. The same-author, unreviewed companion JC and K2P manuscripts fix the semi-directed convention, the cycle–theta topology grammar, restoration, and coherent probes [19, 18]. We reuse that model-independent graph framework but rebuild each K3P algebraic layer.

There is also a sharp warning outside the theorem class. A same-author, unreviewed companion manuscript gives an exact K3P collision between a three-leaf tree and a level-2 double-theta network, whose fixed theta map has ambient normalized rank 15 [17]. Thus three-taxon pointwise blob detection fails in general. That theta topology is not even weakly tree-child, so it does not locate the boundary of the strong theorem. The sharpness construction in section 13 does: ambiguity already becomes full-dimensional in the weakly but not strongly tree-child class.

Our main result gives the complete K3P containment classification within the class of binary standard semi-directed strongly tree-child level-2 networks: regular full-dimensional containment is exactly labelled topology equivalence modulo ordinary triangle redirection, both on the principal positive domain and in strict continuous time. Thus the topology quotient agrees with the JC and K2P classifications, although the local triangle geometry changes. Under K3P the three triangle orientations have rank 14 in the normalized 15-dimensional three-leaf Fourier chart. Their common closure is one irreducible quartic hypersurface H_{14}, and the ambiguity is a common smooth germ relative to that hypersurface, not an ambient-open germ.

The proof has six layers.

- **L1.** The principal stochastic and strict continuous-time domains are handled exactly. Reversibility gives root invariance, and an isotropic near-identity factor gives strict physical subdivision.
- **L2.** Bridge topology is transferred noncircularly under a directed containment. A true bridge always has flattening rank at most four; a nonbridge is generically detected by a displayed-tree boundary specialization and an explicit nonzero quartet minor. The missing reverse cut inclusion is then proved inside the strong class by a crossing-split dichotomy and 204 pointwise one-active K3P certificates.
- **L3.** Positive bridge flattenings yield exactly three labelled incidence scales per component–bridge incidence. Explicit analytic normalizers and strict serial splitting give a physical local product chart with no sector-permutation gauge and no holonomy.
- **L4.** Every selected marginal is a three-product physical submersion. A finite semialgebraic-cover argument localizes a global relation to a fixed incoming and completion type, excluding compensation from remote factors.
- **L5.** The bounded K3P residue consists of fourteen four-port directed relation orbits and two pre-lock sink swaps. Nine orbits have exact quartic separators and five have strict directed-rank obstructions. Exact restoration and coherent one-/two-port probes extend the bounded theorem to arbitrary labelled subdivision words.
- **L6.** The common smooth H_{14} germ contextualizes through one labelled contraction map, and simultaneous physical bridge gluing supplies sufficiency. Semialgebraic stratification then gives generic identifiability and a terminating exact reconstruction algorithm.

One correction to an earlier program statement is load-bearing. We do *not* claim that every arbitrary strict K3P noncut has flattening rank greater than four pointwise. The exact statements used here are: true cuts have rank at most four at every strict point; each noncut has rank greater than four generically; and the cut sets of two strong-class networks are equal whenever a source-relative containment exists. The last statement is stronger than generic recovery for the purpose of containment and is proved without assuming a common bridge tree.

The finite portions are computer-assisted, but the proof boundary is explicit. Graphs, directions, incoming roles, repairs, port maps, Fourier transports, exact pullbacks, rank witnesses, restoration parentage, and probe restrictions are all serialized. A separate implementation rebuilds the load-bearing claims, and mutations test the principal failure modes. The reader supplement gives the exact census and theorem-to-certificate crosswalk; full ledgers remain in the reproducibility archive rather than in the prose.

2 Main theorems

We first state the results; definitions appear in section 3. Every edge is strict and every inheritance probability belongs to $(0, 1)$.

Theorem 2.1 (K3P containment classification for strongly tree-child level-2 networks)**.** *Let N, N' be binary standard semi-directed strongly tree-child level-2 networks on the same finite labelled leaf set. Put every edge spectrum in*

$$\mathcal{D}_{3,+} = \{(c, g, t) \in (0, 1)^3 : 1 + c - g - t > 0,\ 1 - c + g - t > 0,\ 1 - c - g + t > 0\}.$$

Then

$$\boxed{N \preceq_{3,+} N' \quad \Longleftrightarrow \quad N \equiv_{\triangle} N' \quad \Longleftrightarrow \quad N \bowtie_{3,+} N'.} \tag{1}$$

In particular, no proper one-sided regular full-dimensional containment occurs in this class. The conclusion is not equality of the complete stochastic images.

Theorem 2.2 (Generic structural identifiability)**.** *For each fixed N in the class of theorem 2.1, there is a proper Zariski-closed set $E_N \subsetneq \mathcal{V}_N$ such that every exact physical tensor $q \in \mathcal{M}_{3,+}(N) \setminus E_N$ determines the labelled standard semi-directed topology of N uniquely modulo $\equiv_{\triangle}$.*

Theorem 2.3 (Exact reconstruction)**.** *There is a terminating exact procedure which, given $q \in \mathcal{M}_{3,+}(N) \setminus E_N$ for an unknown network N in the class, returns the unique structural class $[N]_{\triangle}$. Here exact input means an exact-real representation supporting field operations, polynomial-sign decisions, and real-closed-field quantifier elimination. No bit-complexity, conditioning, or finite-sample guarantee is asserted.*

Corollary 2.4 (Strict continuous-time classification)**.** *The equivalence in (1) and the conclusions of theorems 2.2 and 2.3 hold after replacing $\mathcal{D}_{3,+}$ by*

$$\mathcal{D}_{3,\mathrm{CT}} = \{(c, g, t) \in (0, 1)^3 : c > gt,\ g > ct,\ t > cg\}$$

and replacing $(3, +)$ by $(3, \mathrm{CT})$ in the observational relations.

Theorem 2.5 (Sharpness against weak tree-childness)**.** *For every $n \geq 3$, there are binary level-2 standard semi-directed networks $W_n, W_n' \in W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$ on the same labelled leaf set which are neither labelled-isomorphic nor ordinary-triangle equivalent, and whose strict continuous-time K3P images share a common full-dimensional regular analytic germ of dimension*

$$15 + 6(n - 3) = 6n - 3.$$

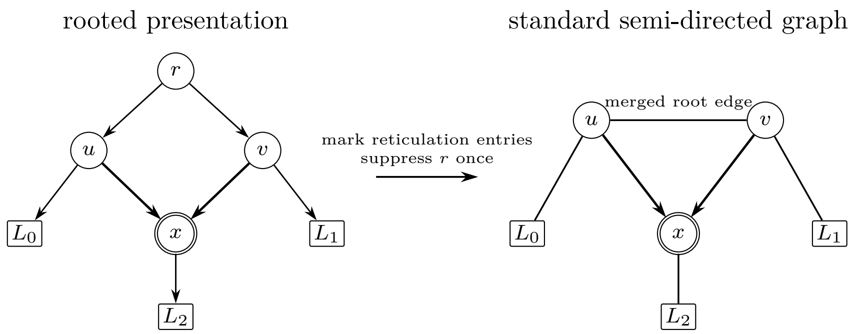


Figure 1: The fixed standard semi-directed convention. The binary root is deleted and its two incident arcs are merged in one step; no further cleanup is performed. The two arrowheads entering the reticulation are retained.

For $n = 3$, both normalized maps have rank 15 throughout a certified rational box containing a unique common equality-slice point.

Proposition 2.6 (Separate outer-class obstruction)**.** *There is a three-leaf tree and a binary semi-directed level-2 double-theta network, the latter outside W_{TC}, for which the K3P tree germ is properly contained in the theta image near a strict continuous-time common point. The normalized ranks are 9 and 15, respectively, and the local theta-parameter collision locus with the tree model has dimension 23. This proposition is not the sharp class-boundary example of theorem 2.5.*

Theorems 2.1–2.5 concern topology and image germs. They neither identify all numerical parameters nor claim that every representative of a triangle class realizes every tensor realized by another representative.

3 Conventions and the K3P model

3.1 Rooted and semi-directed networks

Definition 3.1 (Rooted binary network)**.** *Let X be a finite set of leaf labels. A rooted binary phylogenetic network is a finite directed acyclic graph without parallel arcs in which the root has bidegree $(0, 2)$, every tree vertex has bidegree $(1, 2)$, every reticulation has bidegree $(2, 1)$, and every leaf has bidegree $(1, 0)$ and a unique label in X. Every vertex is reachable from the root. The root is the lowest stable ancestor of the leaves. A rooted presentation is* tree-child *when each internal vertex has a tree vertex or leaf as a child.*

Definition 3.2 (Standard semi-directed convention)**.** *Mark every arc entering a reticulation, undirect every other arc, delete the binary root, and merge its two incident arcs into one edge, retaining at either endpoint exactly the arrowhead present before deletion. We admit the presentation only when this single suppression creates no loop or parallel edge and loses no reticulation arrowhead. No later exhaustive cleanup is part of this operation. The resulting labelled mixed graph is a* standard semi-directed network*. An admissible rooting is a rooted binary network whose one-step suppression is exactly that mixed graph.*

The one-step operation and the retained reticulation arrowheads are shown in figure 1.

Isomorphisms preserve leaf labels, ordinary edges, retained arrowheads, and vertex roles. Restriction to a leaf subset is a distinct operation: after choosing displayed reticulation parents one may prune unlabelled dead ends and suppress unlabelled bivalent vertices. Such cleanup does not enlarge the admissible-rooting set of the original mixed graph. See [21, 15] for broader background on rooted and semi-directed conventions.

Definition 3.3 (Weak and strong tree-childness)**.** *A standard semi-directed network is* weakly tree-child *if at least one admissible rooting is tree-child. It is* strongly tree-child *if it has an admissible rooting and every admissible rooting is tree-child. We denote these classes by* W_{TC} *and* S_{TC}*, respectively.*

Lemma 3.4 (No-omnian criterion)**.** *In the binary fixed convention, a standard semi-directed network with an admissible rooting is strongly tree-child if and only if every tail of a retained reticulation edge is incident with two ordinary edges.*

Proof. If a tail fails the condition, it is either a reticulation with a reticulation child or an ordinary vertex tailing two retained edges. The first case is non-tree-child in every rooting. In the second, moving an admissible root to the tail's unique ordinary incidence produces an admissible rooting in which the tail has two reticulation children. Thus strong tree-childness fails. Conversely, under the incidence condition an ordinary vertex with a reticulation child retains an ordinary child, a reticulation has an ordinary child, and a root with two reticulation children would suppress to a forbidden two-arrowhead edge. Hence every admissible rooting is tree-child. This is the fixed binary specialization of the criterion used in [8, 15]. □

A *bridge* is an edge whose deletion disconnects the underlying graph. A *blob* is a maximal nontrivial biconnected block. A network is *level two* when every blob contains at most two reticulations. Contracting each nontrivial blob while retaining the intervening ordinary branching components gives the labelled reduced *tree of blobs*. When a blob is cut from the network, each incident bridge becomes a labelled boundary *port*. A complete ported factor retains every physical port, including the child port below a path-sink reticulation.

Definition 3.5 (Ordinary triangle equivalence)**.** *An ordinary triangle redirection changes only which vertex of a three-cycle factor incident with three external arms is reticulate and redirects the two arrowheads entering it. Write* $N \equiv_{\triangle} N'$ *when the labelled reduced trees of blobs agree, corresponding nontriangle complete factors are labelled mixed-graph isomorphic, and every remaining pair differs by an ordinary triangle redirection, with coherent boundary transports.*

3.2 Fourier parameterization and physical domains

Identify the states and characters with $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{0, C, G, T\}$, with Klein-four addition. A K3P edge e has Fourier spectrum

$$\widehat{f}_e = (1, c_e, g_e, t_e).$$

Inverse Fourier transformation gives

$$f_e(0) = \frac{1 + c_e + g_e + t_e}{4}, \qquad f_e(C) = \frac{1 + c_e - g_e - t_e}{4},$$
$$f_e(G) = \frac{1 - c_e + g_e - t_e}{4}, \qquad f_e(T) = \frac{1 - c_e - g_e + t_e}{4}. \tag{2}$$

The root state is uniform. At each reticulation r, one parent is chosen with probability $\lambda_r \in (0,1)$ and the other with probability $1-\lambda_r$. A switching σ gives a displayed tree T_σ with positive weight w_σ. For a character assignment $\mathbf{h} = (h_i)_{i\in\mathcal{X}}$,

$$q_N(\mathbf{h}) = \begin{cases} \sum_\sigma w_\sigma \prod_{e\in T_\sigma} \widehat{f}_e(\oplus_{i\in A_e} h_i), & \oplus_{i\in\mathcal{X}} h_i = 0, \\ 0, & \text{otherwise}, \end{cases} \tag{3}$$

where $A_e \mid A_e^c$ is the displayed-tree split. The map is polynomial in edge spectra and inheritance probabilities.

Strict positivity of (2), together with positive nontrivial spectra, is exactly the principal domain $\mathcal{D}_{3,+}$ in theorem 2.1. Symmetric continuous-time K3P rates $(\alpha_C, \alpha_G, \alpha_T) > 0$ give

$$c = e^{-2(\alpha_G+\alpha_T)}, \qquad g = e^{-2(\alpha_C+\alpha_T)}, \qquad t = e^{-2(\alpha_C+\alpha_G)},$$

and are equivalent to the three inequalities defining $\mathcal{D}_{3,\mathrm{CT}}$.

Write $\Theta_{3,+}(N)$ and $\Theta_{3,\mathrm{CT}}(N)$ for the open physical parameter spaces, Φ_N for (3), and $\mathcal{M}_{3,+}(N) = \Phi_N(\Theta_{3,+}(N))$. Put

$$d_N = \max_{\theta\in\Theta_{3,+}(N)} \operatorname{rank} D\Phi_N(\theta).$$

A source parameter is regular when its Jacobian has rank d_N.

3.3 Observational relations

Definition 3.6 (Directed containment and common germ)**.** *Write $N \preceq_{3,+} N'$ if there are a regular source point $\theta_0 \in \Theta_{3,+}(N)$, a connected open neighborhood $U \subseteq \Theta_{3,+}(N)$ on which Φ_N has rank d_N, and a real-analytic physical target section $\sigma : U \to \Theta_{3,+}(N')$ such that*

$$\Phi_N = \Phi_{N'} \circ \sigma \qquad \text{on } U.$$

Write $N \bowtie_{3,+} N'$ when the two physical images contain a common analytic germ that is regular and full-dimensional in both images and has physical analytic sections from both parameter spaces. Define $\preceq_{3,\mathrm{CT}}$ and $\bowtie_{3,\mathrm{CT}}$ by replacing the edge domain with $\mathcal{D}_{3,\mathrm{CT}}$.

These relations are stronger than nonempty image intersection and different from inclusion of complex Zariski closures. The target section in $\preceq_{3,+}$ is physical but need not be target-open or target-regular. This asymmetric quantifier is important in the cut-transfer argument below.

4 Physical domain and topology transfer under containment

This section separates three claims that must not be conflated: pointwise rank at a true cut, generic detection of a fixed noncut, and equality of cut sets under strong-class containment.

Lemma 4.1 (Strict isotropic subdivision)**.** *For $y = (c, g, t) \in \mathcal{D}_{3,+}$, put*

$$B_+(y) = \max\{c, g, t, g+t-c, c+t-g, c+g-t\} < 1.$$

For every $r \in (B_+(y), 1)$,

$$(c, g, t) = (r, r, r) \odot (c/r, g/r, t/r)$$

and both factors lie in $\mathcal{D}_{3,+}$*. In strict continuous time the same holds whenever*

$$r > B_{\mathrm{CT}}(y) := \max\{c, g, t, gt/c, ct/g, cg/t\}.$$

Proof. The first six lower bounds are precisely the upper-coordinate and three inverse-Fourier inequalities for the residual triple. The isotropic factor is strict for $0 < r < 1$. In continuous time, for example, $c/r > (g/r)(t/r)$ is equivalent to $r > gt/c$; the other inequalities are cyclic. All bounds are strictly below one at an interior point. □

Lemma 4.2 (Root movement)**.** *Moving the root to any admissible rooting of the same standard semi-directed mixed graph leaves the K3P physical image and its regular germs unchanged up to analytic physical reparameterization.*

Proof. K3P transition matrices are symmetric and the root distribution is uniform, so reversing ordinary arcs on an old-to-new root path preserves every displayed-tree probability. Use lemma 4.1 at the new root and suppress the old one. If the old root meets a reticulation-parent edge, inspect each switching: the selected parent uses only the serial product, whereas the unselected one-child stem subtends all observed leaves and has total character zero. Every summand of (3) is unchanged after transporting the reversible edge products. More explicitly, write the two parent weights at a reticulation in the recorded parent order as $(\lambda, 1-\lambda)$. Root movement preserves λ when that order is preserved and transports it to $\lambda' = 1 - \lambda$ when the order is reversed; thus the same switching keeps the same coefficient. Strict subdivision supplies analytic physical sections in both directions on a neighborhood, which is the asserted reparameterization. □

For a split $S = A \mid A^c$, order the Fourier flattening by the bridge character $h = \oplus_{i \in A} h_i$. It is block diagonal with four character blocks.

Proposition 4.3 (True-cut rank, pointwise)**.** *At every strict K3P parameter point of every network, if* S *is a bridge split, then*

$$\operatorname{rank} \operatorname{Flat}_S(q) \leq 4.$$

Proof. Condition on the character crossing the bridge. Each of the four character blocks is a positive rank-one outer product of the two side tensors, with the corresponding bridge multiplier in the nonzero blocks. Summing their ranks gives at most four. □

Lemma 4.4 (Displayed-tree witness for a noncut)**.** *If* $S = A \mid A^c$ *is not a bridge split of a strongly tree-child level-2 network, some displayed switching gives a tree that does not display* S*.*

Proof. Color the leaves by the two sides of S, and take the two color hulls in the network's abstract bridge-incidence tree. Disjoint hulls would be separated by a bridge realizing S, contrary to hypothesis. If their intersection contains an edge, that edge is a bridge split crossing S. Every displayed tree retains this bridge split, and two splits of one tree are compatible, so no displayed tree can also display S.

Otherwise the hulls meet in one complete component. Every incident branch there is monochromatic, and minimality of the two hulls supplies at least two branches of each color. The balanced compression argument of lemma 4.7 applies inside this cycle or theta factor and produces a switching with a wrong two-by-two quartet. Extending its local choices arbitrarily over all other reticulations gives the required global displayed tree. □

Proposition 4.5 (Generic noncut recovery)**.** *Fix a standard strongly tree-child level-2 network N and a split S that is not a bridge split. Some* 5*-by-*5 *minor of* $\mathrm{Flat}_S \circ \Phi_N$ *is a nonzero polynomial. Consequently* $\operatorname{rank} \mathrm{Flat}_S(q) > 4$ *on a Zariski-open dense subset of the physical parameter domain.*

Proof. Choose the switching in lemma 4.4. The ordinary-tree quartet criterion supplies $a_0, a_1 \in A$ and $a_2, a_3 \in A^c$ whose restricted displayed tree has a quartet split different from $a_0a_1 \mid a_2a_3$. Work first at the polynomial-map boundary by setting each inheritance probability to zero or one so that only this switching remains, and give every edge an isotropic multiplier strictly between zero and one.

Relabel the displayed quartet split as $01 \mid 23$ and its wrong flattening as $02 \mid 13$. After suppressing its serial paths, write $p_0, p_1, p_2, p_3 \in (0, 1)$ for the four pendant multipliers and $u \in (0, 1)$ for the internal multiplier. In the zero-character block, the rows and columns indexed by 00 and CC give determinant

$$p_0p_1p_2p_3(1 - u^2) > 0.$$

One positive entry from each of the other three character blocks augments this to a nonzero 5-by-5 minor. Assigning character zero to every unselected leaf identifies it with a minor of $\mathrm{Flat}_S \circ \Phi_N$. Thus that network minor has a nonzero boundary evaluation and is not the zero polynomial. By continuity it remains nonzero when every inheritance probability is moved sufficiently close to its selected endpoint but into $(0, 1)$. The isotropic edge points are strict in both $\mathcal{D}_{3,+}$ and $\mathcal{D}_{3,\mathrm{CT}}$. Hence a strict physical evaluation exists, and the zero set of the minor has empty interior in the open K3P parameter domain. □

Corollary 4.6 (The easy directed cut inclusion)**.** *If $N \preceq_{3,+} N'$, then every target bridge split is a source bridge split:*

$$\mathrm{Cut}(N') \subseteq \mathrm{Cut}(N).$$

Proof. For a target bridge, all 5-minors vanish identically after composition with the physical target section. They therefore vanish on the source-open parameter neighborhood. If the split were a source noncut, proposition 4.5 would make one pullback a nonzero polynomial, which cannot vanish on an open set. □

The reverse inclusion needs genuinely K3P pointwise information, but only for a finite one-active universe.

Lemma 4.7 (Balanced noncut compression)**.** *Let a complete strongly tree-child cycle or theta factor carry a two-color boundary assignment with at least two actual labels of each color and no bridge realizing the color split. There is a restriction which retains the primitive core, one complete* *minimum strong repair, every path-sink child, and at least two actual labels of each color. All required unselected roles remain as zero-character completion ports. The restricted coloring remains noncut, some displayed switching fails it, and two actual labels of each color give a wrong quartet in the finite one-active universe of lemma 4.8.*

Proof. First retain a rigid support Q: every path-sink child and one occupied port on every segment of one complete minimum repair. The core table in section 8 gives $|Q| \leq 4$, and suppressing unselected ordinary subdivisions leaves the primitive cycle or theta core. A nonroot incoming boundary, when present, is retained in addition; because it is already an actual colored label, the number of further representatives needed to reach two of each color decreases by one. Thus the compulsory support plus the four-active representatives still has at most eight ports. On each directed segment, read the colors as a word and decompose it into maximal constant runs.

If some segment contains three consecutive runs c, d, c, retain one actual label from each of those runs and, if necessary, one further d-label from the same or another segment. In every switching the path between the two outer c-attachments contains the middle d-attachment. No tree edge can separate the complete color classes: a segment edge separates the outer c-ports or retains the middle d with them, a pendant edge isolates one port, and another d-port exists. This is a direct all-switching obstruction.

Otherwise every segment has at most two runs. Select actual labels from at most two runs of each color, preferring labels already in Q, until two labels of each color are retained. If a color is represented by only one selected label, retain a second actual label in the same run when possible, and otherwise from another run or segment. Its primitive blob and all compulsory child roles remain; no internal blob edge is a bridge, and a pendant arm cannot separate a color class containing two labels. Thus the restricted split is still noncut.

It remains to pass from this bounded restriction to the exact four-active palette. Retain one actual label from each selected run, preserving every occupied segment and fixed child role. Each segment word is now one of

$$(), \quad (0), \quad (1), \quad (0,1), \quad (1,0).$$

Double a singleton run when needed to keep two actual labels of its color. Any split displayed by every switching before this reduction would remain so after leaf restriction. The independent finite switching replay has zero survivors in this direct-or-singleton-doubled palette, so in this second case some switching fails the color split. In either case choose a failing switching. The ordinary-tree quartet criterion then selects two retained actual labels of each color whose quartet fails that split. Every other repair or path-sink role is retained at character zero, so the result is one of the graph-derived one-active directions rather than a pruned topology.

The exact censuses are reproducibility checksums, not premises. A standalone word program exhausts all (808,642) balanced distributions with four through eight active ports and partitions them into (229,988) three-run obstructions, (544,350) direct-palette reductions, and (34,304) singleton-doubled reductions. A no-import implementation independently constructs the five primitive cores and tests (379,742) valid compressed presentations, with zero all-switching survivors. Finally, five cores give 72 active-labelled records and 216 split entries; 12 are displayed by every switching, leaving the 204 wrong directions certified in lemma 4.8. □

Lemma 4.8 (One-active K3P obstruction)**.** *For a balanced two-color split across one active strongly tree-child level-2 component, primitive-core and word compression produce exactly 216 labelled split entries. Exactly 12 are displayed by every switching. For each of the remaining 204 normalized wrong-split directions, every strict $\mathcal{D}_{3,+}$ tensor has flattening rank greater than four.*

Exact finite proof. The active model-independent graph producer derives five primitive core templates, 72 active-labelled records, and three 2-by-2 splits per record from literal directed graphs, root suppression, switching choices, and descendant masks. Its fresh certificate is required to agree byte for byte with the exact topology input consumed by the K3P replay. Starting from those independently regenerated switching signatures, that replay recomputes the displayed-by-all flags: it removes 12 entries and gives 204 distinct labelled direction keys, without quotienting by an automorphism. The five-core theorem and the compression proof are handwritten above; the 72-row labelled table is also graph-derived active evidence, not a K3P algebraic conclusion.

For 180 directions one selected block minor has a strictly signed Bernstein expansion on the strict K3P simplex. Twelve more are excluded by positive signed combinations of two minors. Ten use cyclic six-minor identities. The last two use two independently checked cyclic elimination certificates, one obtained by an exact labelled transport. The disjoint partition is

$$204 = 180 + 12 + 10 + 1 + 1.$$

If the total flattening had rank at most four, its four positive character blocks would each have rank one, forcing all selected minor equations to vanish. The corresponding signed or cyclic certificate contradicts that system throughout $\mathcal{D}_{3,+}$. A separately implemented verifier rebuilds all 204 normalized directions and all exact K3P arithmetic from that table. The graph producer, byte comparison, and coherent topology/mask mutations are separate active gates. The supplement records this implementation boundary. □

Theorem 4.9 (Strong-class cut-set equality under containment). *If $N \preceq_{3,+} N'$ for networks in the class of theorem 2.1, then*

$$\mathrm{Cut}(N) = \mathrm{Cut}(N').$$

Hence the labelled abstract bridge-incidence trees agree. This conclusion does not yet decide whether a trivalent internal component is ordinary or is the contraction of a three-boundary cycle; that decoration is recovered in lemma 10.1.

Proof. By corollary 4.6, only $\mathrm{Cut}(N) \subseteq \mathrm{Cut}(N')$ remains. Suppose a nontrivial source bridge split $S = A \mid B$ is absent from the target; a singleton split is pendant in every standard network and cannot be lost, so $|A|, |B| \geq 2$. Color target leaves by A, B, and in the target reduced tree let H_A, H_B be the two color hulls. They intersect, because an edge separating disjoint hulls would realize S as a target bridge.

If $H_A \cap H_B$ contains an edge, its target bridge split $R = C \mid D$ has a leaf in each of $A \cap C, A \cap D, B \cap C, B \cap D$; thus R crosses S. But corollary 4.6 makes R a source bridge. Two edge splits of one tree are compatible, so source bridge splits R, S cannot cross. This excludes the two-active alternative without assuming a common bridge tree.

Otherwise $H_A \cap H_B = \{v\}$ is one target component. Every branch of the target reduced tree at v is monochromatic. Indeed, if one branch contained both colors, then, because v lies in each color hull, the edge entering that branch would belong to both H_A and H_B, contrary to their intersection being only v. Moreover, membership of v in each minimal color hull forces at least two incident branches of each color: a labelled tree leaf has only one color and cannot lie in both hulls, so v is internal in each hull. Thus v has at least four relevant branches. A binary ordinary tree vertex has degree three, so v represents a nontrivial central blob.

Apply lemma 4.7 to this central blob. It supplies actual labels $a_1, a_2 \in A$ and $b_1, b_2 \in B$ whose induced quartet fails to display $a_1a_2 \mid b_1b_2$, while the completion roles are retained at zero character. This is one of the 204 graph-derived labelled wrong-split directions in lemma 4.8.

It remains to check that suppressing the material between the four selected labels and the central blob stays inside the strict K3P domain. A nonempty ordinary serial path is a convolution of strictly positive $\mathbb{Z}_2^2$ transition distributions, hence is a strict K3P edge. Every component strictly between v and one selected label has only two retained boundaries. Marginalizing such a noncentral blob gives

$$M_{\text{eff}} = \sum_s w_s M_s, \qquad w_s > 0, \qquad \sum_s w_s = 1,$$

where M_s is the strict K3P path matrix of switching s. Each w_s is a product of local factors λ_r or $1-\lambda_r$, so positivity and normalization follow from summing all local switches. K3P matrices form a linear family, so M_{eff} is K3P; its transition entries are positive convex combinations of positive entries and its three nontrivial Fourier eigenvalues are convex combinations of numbers in $(0,1)$. It is therefore a strict $\mathcal{D}_{3,+}$ edge. Chains of these two-boundary mixtures and ordinary paths remain strict by convolution. Retained inheritance probabilities are λ or $1-\lambda$, still in $(0,1)$. Thus the compressed target quartet lies at a strict physical point of the 204-direction theorem; no target regularity or target-open marginal is being assumed.

Apply the same four-leaf marginal to the containment identity. In Fourier coordinates this sets every omitted leaf character to zero. On the source side the original bridge S survives because two selected labels lie on each side, so proposition 4.3 gives rank at most four. On the identical target tensor, lemma 4.8 gives rank greater than four at every strict target point. This contradiction proves the reverse inclusion. The compatible split system determines the labelled abstract bridge-incidence tree. As usual for edge-split recovery, it does not by itself decorate a trivalent vertex as an ordinary branching component or a three-boundary blob. □

Remark 4.10 (Exact claim boundary). Theorem 4.9 is a directional theorem for two networks in the strong class. It does not promote proposition 4.5 to a universal pointwise converse for arbitrary K3P networks. No later proof uses such a converse.

5 Three-leaf K3P geometry

Normalize $q_{000} = 1$ and order the sixteen conservation-supported three-leaf coordinates as

$$\begin{aligned}(000, 0CC, 0GG, 0TT, C0C, CC0, CGT, CTG,\\ G0G, GCT, GG0, GTC, T0T, TCG, TGC, TT0).\end{aligned} \tag{4}$$

The resulting affine chart has dimension 15.

5.1 Trees and ordinary sunlets

A three-leaf tree with pendant spectra a, b, c has

$$q_{xyz} = a_x b_y c_z \qquad (x+y+z=0).$$

For the ordinary sunlet with reticulation adjacent to the third leaf, use the exact map

$$q_{xyz} = a_x b_y c_z \big[\lambda f_y d_z + (1-\lambda) f_x e_z\big], \qquad x+y+z=0. \tag{5}$$

The other orientations are obtained by labelled port transport.

For two triples $L=(u,v,w)$ and $R=(u',v',w')$, write $I(L;R) = q_u q_v q_w - q_{u'} q_{v'} q_{w'}$. Define the six cubic circuits

$$\begin{aligned}
I_1 &= I((000, CGT, GTC); (0TT, C0C, GG0)),\\
I_2 &= I((000, CTG, TGC); (0GG, C0C, TT0)),\\
I_3 &= I((000, GCT, TGC); (0CC, GG0, T0T)),\\
I_4 &= I((000, GTC, TCG); (0CC, G0G, TT0)),\\
I_5 &= I((000, CTG, GCT); (0TT, CC0, G0G)),\\
I_6 &= I((000, CGT, TCG); (0GG, CC0, T0T)).
\end{aligned}$$

Brits et al. [4, Lemma 4.2] prove tree–sunlet disjointness on their restricted stochastic, nonidentity, positive-definite parameter space Θ_0, with inheritance parameters in $(0,1)$, using four tree invariants. The following symmetric six-circuit sum of squares supplies a single pointwise observable on the principal positive domain $\mathcal{D}_{3,+}$ and is the form used in our restoration and probe arguments.

Proposition 5.1 (Tree–ordinary-sunlet separation)**.** *The observable*

$$\mathcal{S} = \sum_{j=1}^{6} I_j^2$$

vanishes identically on the three-leaf K3P tree model and is strictly positive on every ordinary three-sunlet with strict $\mathcal{D}_{3,+}$ edges and $0<\lambda<1$. Leaf permutations cover all three orientations.

Proof. Tree monomials make every I_j vanish. On (5), each pullback has the following literal factorization. Put $\mu = \lambda(1-\lambda)$ and

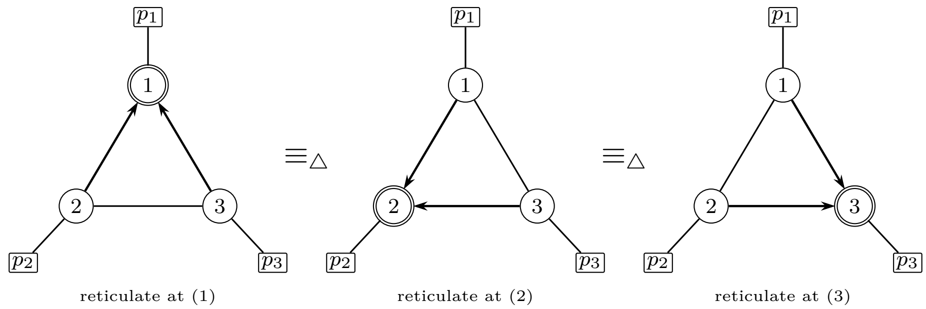
$$\begin{aligned}
&M_1 = a_C a_G b_G b_T c_C c_T, \quad M_2 = a_C a_T b_G b_T c_C c_G, \quad M_3 = a_G a_T b_C b_G c_C c_T,\\
&M_4 = a_G a_T b_C b_T c_C c_G, \quad M_5 = a_C a_G b_C b_T c_G c_T, \quad M_6 = a_C a_T b_C b_G c_G c_T.
\end{aligned}$$

Then exact expansion of the displayed map gives

$$\begin{aligned}
I_1 &= \mu M_1 (f_C f_T - f_G)(d_C e_T - d_T e_C f_G),\\
I_2 &= \mu M_2 (f_C f_G - f_T)(d_C e_G - d_G e_C f_T),\\
I_3 &= -\mu M_3 (f_C f_T - f_G)(d_C e_T f_G - d_T e_C),\\
I_4 &= -\mu M_4 (f_C f_G - f_T)(d_C e_G f_T - d_G e_C),\\
I_5 &= -\mu M_5 (f_C - f_G f_T)(d_G e_T - d_T e_G f_C),\\
I_6 &= \mu M_6 (f_C - f_G f_T)(d_G e_T f_C - d_T e_G).
\end{aligned}$$

All arm and inheritance factors are positive. If $f_C f_T - f_G \neq 0$, simultaneous vanishing of I_1, I_3 forces $f_G^2 = 1$; similarly I_2, I_4 force $f_T^2 = 1$, and I_5, I_6 force $f_C^2 = 1$. These conclusions are impossible for strict edges. If instead all three composition margins vanish, then

$$f_G = f_C f_T, \qquad f_T = f_C f_G, \qquad f_C = f_G f_T,$$



$$\boxed{\overline{\operatorname{im}\Phi_1} = \overline{\operatorname{im}\Phi_2} = \overline{\operatorname{im}\Phi_3} = H_{14}, \quad \operatorname{rank}_{\text{gen}} = 14}$$

Figure 2: The three labelled ordinary-triangle orientations. The external ports stay fixed while the two retained arrowheads are redirected to a different triangle vertex. Their normalized K3P image closures are the same irreducible hypersurface H_{14}, and they share a strict smooth rank-14 germ.

so their product gives $p = p^2$ for $p = f_C f_G f_T \in (0, 1)$, again impossible. Therefore at least one circuit is nonzero. A self-describing exact certificate independently expands these six eight-term identities from the literal map; no edge-letter permutation is used. □

This separator is pointwise on the complete three-leaf maps; it does not classify a pruned mixed graph by topology alone.

5.2 The triangle hypersurface

Define the eight-term quartic

$$\begin{aligned} F_{H_{14}} = {} & q_{000}q_{CGT}q_{GTC}q_{TCG} - q_{000}q_{CTG}q_{GCT}q_{TGC} \\ & - q_{0CC}q_{CGT}q_{G0G}q_{TT0} + q_{0CC}q_{CTG}q_{GG0}q_{T0T} \\ & + q_{0GG}q_{C0C}q_{GCT}q_{TT0} - q_{0GG}q_{CC0}q_{GTC}q_{T0T} \\ & - q_{0TT}q_{C0C}q_{GG0}q_{TCG} + q_{0TT}q_{CC0}q_{G0G}q_{TGC}. \end{aligned} \tag{6}$$

Let $H_{14} = Z(F_{H_{14}}) \subset \mathbb{A}^{15}$.

Theorem 5.2 (Ordinary-triangle geometry)**.** *The three labelled ordinary-triangle orientation maps have generic normalized rank* 14*, have the same irreducible image closure* H_{14}*, and share a strict continuous-time smooth rank-*14 *analytic germ. They do not contain an ambient-open* 15*-dimensional germ.*

The three labelled redirections and their common relative hypersurface germ are summarized in figure 2.

Proof. Exact substitution of all three transported versions of (5) annihilates (6). Give edges a, b, c, d, e the isotropic spectrum $(1/2, 1/2, 1/2)$, give f the spectrum $(1/3, 1/3, 1/3)$, and set $\lambda = 1/2$. Every orientation then maps to the same tensor

$$q_0 = \left(1, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{1}{48}, \frac{1}{48}, \frac{1}{12}, \frac{1}{48}, \frac{1}{12}, \frac{1}{48}, \frac{1}{12}, \frac{1}{48}, \frac{1}{48}, \frac{1}{12}\right).$$

All edges are strict continuous-time. Exact 14-minors at the three preimages have determinants

$$-\frac{1}{760840571584512}, \qquad -\frac{1}{760840571584512}, \qquad \frac{1}{760840571584512}.$$

The quartic gradient at q_0 is nonzero; its six nonzero normalized components have absolute value $1/6912$. Thus H_{14} is smooth and has dimension 14 at q_0. Every orientation differential has rank 14 and lies in the 14-dimensional tangent kernel of $dF_{H_{14}}$, so it submerses onto a relative neighborhood in H_{14}. Intersect the three neighborhoods and apply the constant-rank theorem to obtain physical analytic sections.

For irreducibility, regard the normalized quartic as linear in q_{0CC}. Its coefficient is the primitive disjoint-support binomial

$$-q_{CGT}q_{G0G}q_{TT0} + q_{CTG}q_{GG0}q_{T0T}.$$

The exponent difference is primitive, so this coefficient is irreducible. A hand-checkable specialization is obtained by setting $q_{000} = q_{CTG} = q_{GCT} = q_{TGC} = 1$ and every other coefficient-ring variable occurring in (6) to zero. The coefficient then vanishes while the remainder equals -1; hence the coefficient does not divide the remainder. Gauss's lemma proves that $F_{H_{14}}$ is irreducible. Since every orientation is contained in this hypersurface and contains a relative neighborhood of its smooth locus, each image closure is exactly H_{14}. None is ambient-open in $\mathbb{A}^{15}$. □

For comparison, the normalized three-leaf tree map has rank 9; one exact minor is 8192/28940625. The outer-class double-theta map in section 14 has rank 15. The rank-14 triangle ambiguity is therefore intermediate and must be contextualized relative to H_{14}, not to the ambient tensor chart.

Lemma 5.3 (Contextual H_{14} germ)**.** *Replace one ordinary triangle orientation by another inside an otherwise unchanged labelled context, possibly reconnecting two triangle terminals inside a theta factor. The two complete network maps share a common germ that is full-dimensional relative to both images.*

Proof. Let $U \subset H_{14}$ be the common relative germ from theorem 5.2, C a physical chart of the unchanged context, and write the common labelled multilinear contraction as

$$\Psi(u, c)(g) = \sum_{\mathbf{h} \in \mathbb{Z}_2 \times \mathbb{Z}_2^3} C(g; \mathbf{h}, c)u(\mathbf{h}).$$

Choose one point of $U \times C$ where $D\Psi$ has its generic maximal rank and shrink to a constant-rank neighborhood. Physical sections of each triangle orientation through U give the same lower rank for each contextual map; direct factorization of each map through Ψ gives the reverse inequality. A constant-rank section of Ψ, followed by the orientation sections, yields one common full-dimensional contextual germ. No tensor-product independence among the three terminals is assumed. □

6 The three-sector bridge fibre

Cut every bridge of the labelled reduced tree. Its component-incidence graph is a tree T. For a component v, let P_v be its normalized Fourier boundary tensor, including one character h_e for every incident bridge, with $P_v(0) = 1$. For $e = uv$, write

$$k_e(0) = 1, \qquad k_e(C) = c_e, \quad k_e(G) = g_e, \quad k_e(T) = t_e.$$

Character conservation gives the global contraction

$$Q(\mathbf{h}) = \prod_{v \in V(T)} P_v(\mathbf{h}|_v) \prod_{e \in E(T)} k_e(h_e). \tag{7}$$

Theorem 6.1 (Complete positive three-sector bridge fibre)**.** *On the positive normalized K3P component-tensor locus, the complete fibre of* (7) *is exactly*

$$P_v(\mathbf{h}) \longmapsto P_v(\mathbf{h}) \prod_{e \ni v} \prod_{s \in \{C,G,T\}} (a^s_{v,e})^{\mathbf{1}[h_e = s]}, \tag{8}$$

$$k_e(s) \longmapsto \frac{k_e(s)}{a^s_{u,e} a^s_{v,e}} \qquad (s = C, G, T), \tag{9}$$

with positive incidence scales and $a^0_{v,e} = 1$. There is no additional positive continuous or discrete gauge, the action is free on every retained component, and the bridge tree has no holonomy. On physical networks the fibre is the intersection of this orbit with all local and edge inequalities.

Proof. Cut one bridge and fix $s \in \{C, G, T\}$. Its Fourier block is a positive rank-one matrix, whose factorization is unique up to one positive scalar. All-zero normalization fixes the zero-sector scalar. Unlike K2P, the three nonzero characters are observably labelled and independent; no equality or permutation is imposed. Peeling leaves of T sector by sector assigns one scale to every incidence and gives (8)–(9). Direct cancellation proves the converse, while the tree structure leaves no cycle on which a holonomy could circulate.

Call a retained component *marked* if it carries a retained physical leaf boundary block. For each incident bridge and nonzero sector, that block supplies the compensating character in a conservation-supported one-character anchor. Call the component *unmarked* otherwise. Every unmarked retained component in the strong class has bridge degree at least three: the only possible degree-two theta stabilizer is the two-boundary $K_4 - e$ factor, which has no tree-child rooting.

For freeness, a marked component has one-character anchors with identity exponent matrix. For an unmarked degree-d component, $d \geq 3$, use pair anchors

$$(1, 2), (1, 3), (2, 3), (1, 4), \dots, (1, d).$$

In one sector the leading 3-by-3 exponent block has determinant -2, and every later row introduces one new coordinate. Its positive inverse is

$$a_1 = \sqrt{\frac{r_{12} r_{13}}{r_{23}}}, \quad a_2 = \sqrt{\frac{r_{12} r_{23}}{r_{13}}}, \quad a_3 = \sqrt{\frac{r_{13} r_{23}}{r_{12}}}, \quad a_k = \frac{r_{1k}}{a_1}. \tag{10}$$

Three block-diagonal copies have rank $3d$ and leading determinant $(-2)^3 = -8$. Positivity makes the slice analytic. The excluded unmarked degree-two stabilizer cannot occur in a retained strong-class factor: the two-boundary $K_4 - e$ theta has no tree-child rooting. Finally, a distinct-spectrum anchor such as $(q_{CC}, q_{GG}, q_{TT}) = (1/4, 1/9, 1/16)$ excludes a hidden permutation of the fixed state labels. $\square$

Lemma 6.2 (Physical local product chart)**.** *Near every regular physical point, the quotient by the action in theorem 6.1 is a physical local product chart, simultaneously on all bridges, in both $\mathcal{D}_{3,+}$ and $\mathcal{D}_{3,\mathrm{CT}}$.*

Proof. For one bridge spectrum y, choose two isotropic endpoint factors near the identity and a compensating residual factor, using lemma 4.1. All six endpoint sector coordinates may vary independently in a sufficiently small neighborhood while the residual stays strict. Do this for finitely many bridges and intersect the neighborhoods. The analytic normalizers extract local projective component tensors and effective bridge coordinates from the global tensor, while contraction is their local inverse. The same argument uses B_{CT} on the continuous-time cone. □

For a complete ported factor H, quotient its normalized tensor by one positive C, G, T scale per port and write $\overline{\Phi}_H$ for a local analytic slice map. Local directed containment is defined exactly as in definition 3.6, using a source-open regular germ and a physical target section. Different positive anchor charts give equivalent definitions by theorem 6.1.

7 Marginal localization

Lemma 7.1 (Triple-product marginal submersion)**.** *For every $m \geq 1$,*

$$((c_i, g_i, t_i))_{i=1}^m \longmapsto \left(\prod_i c_i, \prod_i g_i, \prod_i t_i \right)$$

maps $\mathcal{D}_{3,+}^m$ onto $\mathcal{D}_{3,+}$, has differential rank three, and has local physical analytic sections. The analogous statement holds on $\mathcal{D}_{3,\mathrm{CT}}$.

Proof. Serial composition of strict K3P transition distributions is group convolution and therefore remains strict; Fourier transformation converts it to coordinatewise multiplication. For surjectivity, choose the first $m-1$ factors isotropic and sufficiently near the identity, and put the strict residual on the final edge, as in lemma 4.1. The three Jacobian rows have disjoint positive supports; derivatives with respect to (c_1, g_1, t_1) give a nonzero diagonal minor. Continuous-time surjectivity follows by using B_{CT}, or by positive coordinate power roots. □

The exact graph-derived descriptor records more than ordinary serial paths. For a retained character assignment and switching σ, an edge sees the group sum below its displayed descendant mask. The complete visible signature is the array of those C, G, T values over every switching and retained assignment. Split complements agree because the total group sum is zero. Edges with the same nonzero signature enter only through their three products; zero-signature edges are invisible. A retained inheritance coordinate is transported as λ or $1 - \lambda$ according to the recorded parent order; a tensor-invisible reticulation disappears after its two switching weights sum to one.

Proposition 7.2 (Complete marginal descriptor)**.** *Every rigid-support, restoration-prefix, one-port, and two-port restriction used in the proof factors into triple-product maps from lemma 7.1, coordinate projections and permutations, and recorded parent-order complements. It is physically onto its effective selected domain, is a submersion, and has local physical analytic sections.*

Proof. Set omitted leaf characters to zero in (3) and group full switchings by their retained switching. Identical complete signatures have identical sector exponents in every monomial and therefore collapse to one triple product. Tensor-invisible reticulations contribute $\lambda + (1-\lambda) = 1$; every other inheritance coordinate is transported exactly. Apply lemma 7.1 independently to each visible class. □

Lemma 7.3 (Finite semialgebraic cover)**.** *Let a nonempty relatively open subset of a smooth d-dimensional semialgebraic manifold be covered by finitely many semialgebraic sets. One member of the cover contains a nonempty relatively open subset.*

Proof. A semialgebraic subset of a smooth d-manifold has dimension d exactly when it has nonempty relative interior. Finite-union dimension and the frontier-dimension inequality therefore give the claim [3, Propositions 2.8.5(i) and 2.8.13]. □

Proposition 7.4 (Physical localization and no compensation)**.** *Suppose $N \preceq_{3,+} N'$. For each pair of corresponding complete ported factors H, H', some fixed target incoming and completion type contains a source-relative full-dimensional regular germ of the projective physical K3P model of H. Parameters in other factors cannot compensate for a local separator.*

Proof. By theorem 4.9, the labelled abstract bridge-incidence trees agree. Shrink the regular source neighborhood around a nonzero maximal minor. The physical product chart of lemma 6.2 contains a product box of local factor germs and bridge intervals. Fix all coordinates except one focal factor. Positive bridge factorization and the analytic normalizers extract its projective incidence orbit intrinsically from the global tensor.

Only finitely many target incoming and completion types occur in the primitive grammar. Let R_H be the rank-drop locus of the projective focal map. Its complement is a nonempty semialgebraic open set: a maximal focal Jacobian minor is nonidentically zero, and the physical product chart identifies its rank with the focal summand of the global rank. Work in a small maximal-rank source box U in that complement.

For a fixed target type τ, let Ψ_τ be its projective physical factor map, including the effective marginal coordinates supplied by proposition 7.2, and form the physical incidence correspondence

$$Z_\tau = \{(u, \eta) \in U \times \Theta_\tau : \overline{\Phi}_H(u) = \Psi_\tau(\eta)\}.$$

This is semialgebraic. The original global analytic target section, followed by intrinsic factor extraction, shows that the finitely many projections $\mathrm{pr}_U Z_\tau$ cover U; it is used here only to establish coverage, not as the desired fixed-type section. By Tarski–Seidenberg each projection is semialgebraic, so lemma 7.3 gives a type τ whose projection contains a relatively open source subgerm U_H.

Take a finite semialgebraic real-analytic constant-rank stratification of Z_τ, compatible with the projection to U. The image of a stratum on which that projection has rank below $\dim U$ has dimension below $\dim U$. Because finitely many such images cannot contain the relatively open set just obtained, some incidence stratum projects with rank $\dim U$. Shrink about a physical point over that open image and call the resulting open source-parameter subbox U_H. The analytic constant-rank theorem supplies a physical analytic section $u \mapsto (u, \sigma_{H,\tau}(u))$ of the incidence projection on U_H. Hence

$$\overline{\Phi}_H = \Psi_\tau \circ \sigma_{H,\tau}$$

on U_H, exactly the fixed-type local directed relation. If collapsed serial classes must be expanded into physical edge coordinates, compose with the local physical analytic section in proposition 7.2. Intrinsic extraction is independent of remote target coordinates, so a polynomial or rank obstruction on the focal orbit cannot be cancelled elsewhere. □

Lemma 7.5 (Fixed-full restoration)**.** *Fix actual networks and a full relation $N \preceq_{3,+} N'$. Fix one omitted physical label, its source insertion edge, target attachment, and boundary transport. Marginalizing this same relation gives one enumerated child relation with source-open restriction. Exact separation of every such child therefore excludes the fixed full parent.*

Proof. Marginalization sets the restored character to zero. Pendant factors become one, serial classes multiply coordinatewise, and proposition 7.2 gives a physical source-open image. The actual attachment and transport are one row of the exhaustive child construction. No abstract smaller relation is lifted and no arbitrary target deletion map is inverted. □

8 Primitive factors and bounded classification

8.1 Cycle and theta cores

Move a root-containing factor along an all-tree path to a real labelled incoming boundary, using lemma 4.2. Delete port arms and suppress ordinary bivalent vertices that are neither the local source nor a reticulation sink, recording the removed port-bearing subdivisions as ordered words.

Proposition 8.1 (Cycle–theta primitive theorem)**.** *Every nontrivial complete blob factor in the strong level-2 class has a cycle core or one of four directed theta cores $\theta_0, \theta_1, \theta_2, \theta_3$. It is recovered uniquely from the core, one labelled ordered port word on each directed segment, and one child port below every path-sink reticulation.*

Proof. Let r, v, e be the numbers of reticulations, internal vertices, and blob edges after port arms are removed. Summing indegrees gives $e = v + r - 1$, hence

$$\sum_w (\deg_B(w) - 2) = 2e - 2v = 2(r - 1).$$

For $r = 1$, every blob vertex has degree two and the core is a cycle. For $r = 2$, exactly two vertices have degree three; biconnectedness makes the blob three internally disjoint paths between them, hence a theta. The incoming boundary is ordinary: deleting an arc entering a reticulation cannot disconnect its endpoints, because the other parent and their root paths give an alternate connection. Its factor endpoint is therefore the unique local tree source S. A theta pole already uses all three binary incidences in the blob, so S lies inside one pole-to-pole path. Any internal reticulation is a path sink whose unique child is its boundary arm.

At most one pole is reticulate. If both were, the path containing S points toward both poles and cannot contain either pole's unique outgoing side. Those two outgoing sides must therefore lie on the other event-free paths, forcing one complete path from the first pole to the second and the other back from the second to the first. That is a directed cycle.

Suppose exactly one pole R is reticulate and the other T is a tree pole. The second reticulation is an internal sink X. If S and X lie on different paths, the S-path points away from S, the X-path points toward X, and the third path is forced from T to R; this is θ_0. If they share a path, the only acyclic order from T to R is T–S–X–R. The reverse event order would force the two event-free paths in opposite directions and create a directed cycle. Both event-free paths therefore point from T to R, giving θ_1.

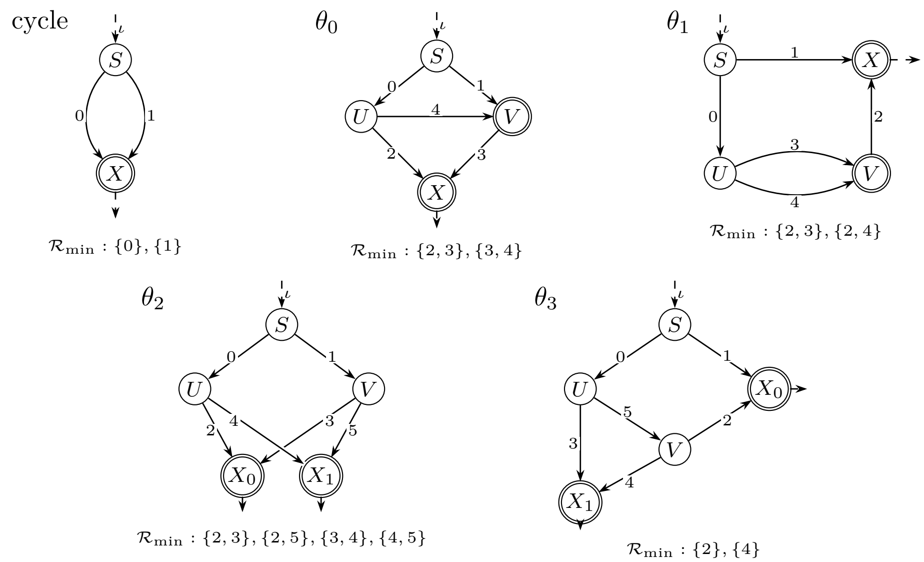


Figure 3: The five primitive complete-factor cores. Segment indices agree with the port-word grammar; ι is the selected incoming boundary. The complete minimum strong repairs $\mathcal{R}_{\min}$ are shown below each core. Single circles mark source or tree roles, double circles mark reticulations, and dashed stubs retain incoming or path-sink child ports.

If both poles are tree vertices, both reticulations are internal sinks. They cannot share one path: the two arrows entering each sink would force S between the sinks, while each of the other two event-free pole-to-pole paths would then have to be directed from one tree pole to the other. Those paths cannot have the same direction, because the receiving pole would have two parents; opposite directions form a directed cycle with the source-to-sink subpaths. Thus the sinks lie on distinct paths. If S lies on the third path, both poles feed both sinks, giving θ_2. Otherwise S shares a path with one sink; read from the pole whose other two sides point outward, acyclicity fixes the order pole–S–sink–pole and every remaining direction, giving θ_3. Reversing unnamed poles or interchanging unnamed sinks yields only the displayed symmetries. Hence no source or sink placement is omitted. This is the directed ported refinement of the two underlying level-2 generators in [16]; the frozen topology replay is a finite regression of this case split, not its premise. □

Figure 3 fixes the pole, segment, incoming-boundary, and minimum-repair notation used in the finite grammar.

The directed segments and minimum occupied sets required by lemma 3.4 are:

core	directed segments in index order	minimum occupied sets
cycle	$S \to X, S \to X$	$\{0\}, \{1\}$
θ_0	$S \to U, S \to V, U \to X, V \to X, U \to V$	$\{2,3\}, \{3,4\}$
θ_1	$S \to U, S \to X, V \to X, U \to V, U \to V$	$\{2,3\}, \{2,4\}$
θ_2	$S \to U, S \to V, U \to X_0, V \to X_0, U \to X_1, V \to X_1$	$\{2,3\}, \{2,5\}, \{3,4\}, \{4,5\}$
θ_3	$S \to U, S \to X_0, V \to X_0, U \to X_1, V \to X_1, U \to V$	$\{2\}, \{4\}$

The corresponding obstruction clauses are $\{0,1\}$ for a cycle; $\{3\}, \{2,4\}$ for θ_0; $\{2\}, \{3,4\}$ for θ_1; $\{2,4\}, \{3,5\}$ for θ_2; and $\{2,4\}$ for θ_3. The displayed sets are precisely their minimum transversals.

For a target core with m_H directed segments, q_H path-sink roles, and r_H repair tags, the exact number of target completions using k selected boundaries and selected-incoming flag ε is

$$C(k,\varepsilon) = \sum_H r_H \sum_{j=0}^{q_H} \binom{q_H}{j} \binom{k-\varepsilon-j+m_H-1}{m_H-1}. \tag{11}$$

The tuples (m_H, q_H, r_H) are $(2,1,1),(5,1,2),(5,1,2),(6,2,4),(6,2,2)$. Thus the four-port target grammar has 831 selected-incoming and 1,983 marginalized-incoming completions.

8.2 The exact four-port residue

Six rigid K3P source supports have normalized ranks

$$20,\ 21,\ 21,\ 21,\ 23,\ 24.$$

After pointwise, quadratic, graph-isomorphism, and ordinary-triangle filters, the only nonterminal four-port residue consists of the following 14 canonical directed orbits. Here the arrow in the rank column is source to target, and "H14" means a transported marginal of (6). The symbol r_j is the zero-based index of the listed minimum occupied set for that core, and a word `p0p1p2p3` records the port relabeling $i \mapsto p_i$. The family labels H21 and Ld record, respectively, a rank-21 nonautomorphic relabeling family and a lower-rank source mapping to a rank-24 target; the suffixes a and b distinguish the two θ_1-repair families.

orbit	source	target	rank	raw	exact certificate
H21-01	θ_0, r_1	θ_0, r_1; `0132`	$21 \to 21$	4	H14 quartic
H21-02	θ_0, r_1	θ_0, r_1; `0213`	$21 \to 21$	4	$11 > 10$ rank
H21-03	θ_0, r_1	θ_0, r_1; `0231`	$21 \to 21$	4	H14 quartic
H21-04	θ_0, r_1	θ_0, r_1; `0312`	$21 \to 21$	4	H14 quartic
H21-05	θ_0, r_1	θ_0, r_1; `0321`	$21 \to 21$	2	H14 quartic
H21-06	θ_0, r_1	θ_0, r_1; `1032`	$21 \to 21$	4	H14 quartic
L20-01	θ_0, r_0	θ_3, r_1; `1203`	$20 \to 24$	2	quartic
L20-02	θ_0, r_0	θ_3, r_1; `1230`	$20 \to 24$	2	$14 > 12$ rank
L21a-01	θ_1, r_0	θ_3, r_1; `0123`	$21 \to 24$	2	quartic
L21a-02	θ_1, r_0	θ_3, r_1; `0132`	$21 \to 24$	2	$11 > 10$ rank
L21b-01	θ_1, r_1	θ_3, r_1; `0123`	$21 \to 24$	2	quartic
L21b-02	θ_1, r_1	θ_3, r_1; `0132`	$21 \to 24$	2	$11 > 10$ rank
L23-01	θ_3, r_0	θ_3, r_1; `0123`	$23 \to 24$	2	$14 > 12$ rank
L23-02	θ_3, r_0	θ_3, r_1; `0132`	$23 \to 24$	2	quartic

The two pre-lock sink swaps are separate from those fourteen orbits. Their common rigid support is θ_3, r_1 with selected incoming port, both sink-child ports, and the repair port on segment 4; the rows below are its two nonautomorphic labelled presentations.

presentation	port relabeling	rank	exact certificate
SINK-0132	`0132`	$24 \to 24$	quartic
SINK-1023	`1023`	$24 \to 24$	quartic

The six H21 rows are the nonisomorphic double cosets of the repaired θ_0 source; the seventh double coset is isomorphic. The complete final post-filter accounting is

$$\boxed{40 = 38 + 2.} \tag{12}$$

The 38 raw records belong to the fourteen named orbits; the two separately tabulated rows are rank-24 θ_3 sink-swap presentations. Neither is a labelled graph isomorphism, and the same exact eight-term quartic has zero target pullback and nonzero strict source value on each.

Theorem 8.2 (Fourteen-orbit classification)**.** *The polynomially separated orbits are*

H21-01, H21-03, H21-04, H21-05, H21-06,
L20-01, L21a-01, L21b-01, L23-02,

and the directed-rank-separated orbits are

H21-02, L20-02, L21a-02, L21b-02, L23-01.

The two sink swaps are polynomially separated. Thus the bounded residue has zero unresolved directions, no new symmetric K3P move, and no proper directed K3P containment.

Exact finite proof. The active exhaustive replay begins with the six source supports, the 2,814 target completions obtained from (11), and all 24 port permutations. It visits all $6 \cdot 2814 \cdot 24 =$ 405,216 directed presentations. Literal graph tests exclude 377,382, leaving 27,834 post-topology presentations. Exact Jacobian minors together with coefficientwise polynomial-vector-field upper bounds exclude 23,054; the quadratic identities exclude 1,968; transported H_{14} marginals exclude 88; and 2,540 enter the independently replayed restoration forest. The remaining presentations comprise 30 labelled isomorphisms, 114 ordinary-triangle presentations, and the 40 records in (12). Thus this accounting is derived from the primitive grammar rather than read from the fourteen-orbit lock.

For a literal target map f, let A be the exact coefficient matrix of the polynomial vector-field identities $J_f V = 0$, and let E evaluate the vector-field coefficients at the recorded strict rational point. Exact linear algebra computes

$$\dim E(\ker A) = \operatorname{rank} \begin{bmatrix} A \\ E \end{bmatrix} - \operatorname{rank}(A).$$

A nonzero evaluation-image minor makes these kernel fields generically independent and hence gives the certified target-rank upper. Sampled Jacobian ranks are used only to supply nonzero lower-rank minors, never as upper bounds. A presentation enters the 23,054-case rank-exclusion class only when its exact source lower rank is strictly greater than that certified target upper.

The separately implemented verifier imports neither the producer nor its graph-atlas module. It reconstructs the grammar, all 4,379 distinct literal maps used by the presentations, the topology tests, every exact nonzero rank minor, every target rank upper bound, and the polynomial pullbacks. It also checks the 2,540 restoration obligations against the active forest by their source, target, and port-permutation data, and derives the double-coset quotient of the 40-record residue. Coherently resealed omission and reclassification mutations are required to fail.

For each of the nine polynomial orbits, an exact characteristic-zero quartic vanishes on the complete target map and has a nonzero value at a strict rational source point. The first five are transported H_{14} marginal quartics; the remaining four are multigraded eight-term quartics. Exact graph and Fourier transports cover every raw member.

For each of the five rank orbits, a selected source projection has a nonzero exact Jacobian minor of rank 11 or 14. The target projection factors rationally through 10 or 12 generators, respectively, giving the strict upper bound. These are directional selected-projection obstructions, not comparisons of total model dimension: the target may have equal or larger total normalized rank while its indicated observable projection has smaller function-field dimension. The H21 upper bound holds after saturation by declared physical factors, all nonzero throughout the strict domain; the other four targets are compressed ordinary-sunlet projections. A source-relative open containment cannot lie inside the lower-dimensional target projection.

The clean-room implementation reconstructs all six sources, all 2,814 target completions, seven H21 double cosets, all 38 coordinate transports, and both sink swaps. The original H21-01 failure is diagnosed as the use of rooted-DAG automorphisms in place of root-suppressed semi-directed mixed-graph automorphisms, compounded by failure to conjugate a base target automorphism into the displayed frame. The corrected mixed-graph transport passes and the historical verifier remains preserved. Exact rows, columns, bodies, source evaluations, and hashes appear in the supplement. □

The bounded result is intentionally direction-sensitive. A target of larger generic dimension is not automatically a containing model, and a polynomial identity is transported through the literal labelled graph rather than attached only to a topology name.

9 Restoration and arbitrary words

The four-port theorem classifies rigid supports, not arbitrary strings of labelled subdivisions along core segments. Two exact extensions close that gap: fixed-full restoration recovers roles omitted by a selected support, and overlapping probes recover every remaining port and its order.

9.1 Fixed-full restoration

The restoration forest is imported only as model-independent graph data: parentage, layer order, source and target identities, dummy roles, insertion sites, and boundary transports. Every K2P polynomial and sector equality is discarded. The K3P proof regenerates the selected maps and certificates from the literal graphs, using lemma 7.5 to connect each child to the original full relation.

Proposition 9.1 (K3P restoration closure)**.** *Every one of the 36,568 first-layer fixed-full restoration children is excluded by an exact K3P observable. Their minimal K3P census is*

certificate	*minimal first layer*	*all-edge replay*
displayed-quartet mismatch	*35,758*	*36,006*
three-sector tree–sunlet sum of squares	*606*	*614*
K3P multihomogeneous quadratic	*148*	*148*
active K3P marginal quartic	*56*	*56*
total	*36,568*	*36,824*

The 32 nodes that were continuations in the inherited graph forest are already terminated by K3P quartics. The full forest is nevertheless replayed as a redundant check: it has 36,824 edges, 256 depth-two edges, 36,792 legacy leaves, maximum depth two, and zero missing children, duplicates, cycles, broken transports, or unresolved rows.

Proof. The 614 rows formerly assigned a two-sector $\mathcal{T}_i$ identity are replaced by the literal K3P three-leaf maps and the six-circuit theorem of proposition 5.1; 606 occur at depth one and 8 in the redundant depth-two replay. The 148 quadratic bodies are regenerated with independent C, G, T variables. The 24 formerly transported K2P quartics and the 32 former continuations use active L20-01/H21-01 K3P quartics, giving the all-edge quartic count $56 = 24 + 32$. Every row is bound to its exact parent transport and direct marginal. Thus no equality such as $C = T$ enters the proof. A producer-independent replay verifies every row, and the fail-closed mutation suite rejects all 20 prescribed corruptions. □

9.2 Coherent probes

The non-four part of the starting universe is derived rather than assumed from the probe contract. Support exhaustion leaves exactly these families: the tree and cycle supports start at three ports; $\theta_0, \theta_1, \theta_3$ enter the complete four-port route; and θ_2 first appears with five ports. A graph-only producer enumerates the elementary tree relation, all (13,440) primitive cycle directions and (536,364) fixed-full cycle restorations, and all (2,946,240) primitive θ_2 directions together with its 576 six-port and 288 seven-port restorations. It tests labelled root-suppressed mixed-graph isomorphism and ordinary triangle redirection directly under the selected-incoming fixed-full-parent convention and derives 133 non-four-port anchors: 117 isomorphisms and 16 triangle relations. A separate no-import implementation rebuilds the primitive grammar and obtains the same canonical structural locators and graph hashes. The anchor-universe suite rejects all 16 attacks, including coherently rebound omissions or corruptions at the raw-parent, one-port, two-port, and extra-terminal interfaces. The frozen contract designates the remaining 43 four-port serialization rows: 26 isomorphisms and 17 triangle relations. The independent full four-port route reconstructs the complete 405,216-presentation atlas and verifies every one of those locators and graph relations. Their disjoint union agrees exactly with the public 176-row contract, giving 143 labelled isomorphisms and 33 ordinary-triangle relations in 39 canonical anchor classes. Thus the contract is only a regression target for non-four completeness; for the four-port part it is a designated serialization input whose every row is checked against the exhaustive active atlas.

The four-port handoff is also checked as a generating set, rather than only row by row. Exact labelled graph-pair transport maps all 144 raw four-port equality presentations to the 26 direct contract seeds. All 1,260 possible first restoration requests then match 161 existing one-port rows, with their relations agreeing exactly; the 12 equality continuations produce 96 second requests, which match 64 existing two-port rows and are all separated there. There is no unmatched equality or restoration obligation. Thus the 43-row four-port serialization is complete for downstream probes even though it is not claimed to be an independent canonicalization of every raw presentation.

There are 176 additional theta presentations whose target incoming boundary has been marginalized and whose selected mixed graphs become abstractly isomorphic only after deleting that role. Promoting all their dummy roles gives 424 fully physical paths, with 56, 176, and 192 at restoration depths one, two, and three. The no-import implementation checks all 984 nonempty prefix equalities. After deleting the restored incoming label and compacting the boundary labels, every terminal path restricts to one of 15 canonical theta seed-pair classes. Transporting the deleted label's two attachment edges then reconciles the paths with 66 existing labelled-isomorphic one-port rows and 66 stored relation classes, with zero unmatched paths. Thus these are downstream one-port root-movement presentations, not omitted anchor parents.

Every physical mixed edge is an attachment site, including pendant arms, reticulation-incoming edges, and the root-suppressed segment.

Proposition 9.2 (One- and two-port coherence)**.** *The complete K3P probe ledgers have the exact censuses*

$$\begin{aligned} 29{,}964 &= 27{,}758 \ \textit{quartet} + 99 \ \textit{tree–sunlet} + 1{,}915 \ \textit{isomorphism} + 192 \ \textit{triangle}, \\ 544{,}571 &= 511{,}266 \ \textit{quartet} + 576 \ \textit{tree–sunlet} + 30{,}969 \ \textit{isomorphism} + 1{,}760 \ \textit{triangle}. \end{aligned}$$

There are 2,107 one-port equality survivors and 32,729 two-port equality survivors, zero unresolved relations, 67,741 exact transports, 4,379 parent restrictions, and zero incoherent triangle choices. One-port restrictions determine the segment of every omitted label; two-port restrictions determine the order of every pair on a common segment. They assemble to one labelled word on every segment, modulo one coherent ordinary triangle redirection. At most nine physical tensor ports are selected.

Proof. Fix one anchor transport. Every non-equality one-port comparison has a direct quartet or K3P tree–sunlet separator, so an equality child locates the new port relative to the rigid support. Every equality two-port comparison restricts to its exact one-port parent and has its reversed marginal in the ledger; this fixes pairwise order. Because the comparisons arise from actual finite words, their orders are transitive and reconstruct a unique total word on each segment.

If a probe subdivides a triangle edge, its appropriate restriction loses the triangle and forces literal orientation agreement. Otherwise it inherits the same stored triangle from the anchor. No isomorphic parent creates a new triangle, and all 1,760 two-port triangle equalities transport the same parent triangle. In addition to the legacy streaming replay, a separately implemented semantic replay reconstructs the literal source and target graphs, marginal restrictions, and incidence- and arrowhead-preserving transports for every one- and two-port row; it also reconstructs each displayed-quartet split and, for every tree–sunlet row, all six row-specific circuit pullbacks. □

Combining the bounded theorem, restoration, and probes gives the exact local statement needed globally.

Theorem 9.3 (Complete local factor classification)**.** *Let H, H' be complete ported cycle/theta factors induced by networks in the strong level-2 class, with arbitrary admissible labelled subdivision words and independently chosen incoming presentations. Then*

$$H \preceq_{3,+} H' \quad \Longleftrightarrow \quad H \cong H' \ \textit{as labelled mixed graphs} \quad \textit{or} \quad H \equiv_{\triangle} H'.$$

No proper one-sided projective local containment occurs.

Proof. Marginal localization fixes one target completion type. The rigid support is handled by theorem 8.2 and the earlier exact terminal filters. Proposition 9.1 discharges every fixed-full parent, and proposition 9.2 reconstructs arbitrary words and the global triangle transport. The converse is immediate for labelled isomorphism and follows from the relative common H_{14} germ for an ordinary triangle. $\square$

10 Global classification

We now prove the main equivalence.

Lemma 10.1 (Trivalent component decoration). *Assume $N \preceq_{3,+} N'$ in the strong level-2 class. After the abstract labelled bridge-incidence tree has been recovered, corresponding trivalent internal components are either both ordinary branching components or both ordinary three-cycle factors.*

Proof. By theorem 4.9 and lemma 6.2, corresponding component positions and their three labelled incidence orbits are intrinsic. Fix every source product coordinate away from one trivalent component and compose this slice with the physical analytic target section. The extracted three-port tensors agree up to positive labelled incidence scaling

$$q_{xyz} \longmapsto A_x B_y C_z q_{xyz}, \qquad A_x, B_y, C_z > 0.$$

In each cubic circuit I_j of proposition 5.1, the two monomials have the same character multiset at each port. Incidence scaling therefore multiplies I_j by one positive monomial and preserves whether it vanishes. All six circuits vanish on an ordinary three-leaf tree, whereas at least one is nonzero at every strict ordinary sunlet point. Hence an ordinary component cannot correspond in either direction to a three-cycle component.

There is no other nontrivial trivalent component in this class. Indeed, the minimum strong repairs in section 8, together with the incoming and path-sink child ports, give every theta factor at least four physical boundaries. A nontrivial three-boundary blob is therefore precisely an ordinary cycle. This proves the decoration claim. $\square$

Proof of theorem 2.1: necessity. Assume $N \preceq_{3,+} N'$. The noncircular cut-transfer theorem theorem 4.9 gives equality of the labelled bridge splits and hence the same labelled abstract bridge-incidence tree. The complete bridge fibre and its physical product chart extract corresponding projective local factors with exactly three incidence gauges and no holonomy. The trivalent decoration lemma now distinguishes every ordinary branching component from every three-boundary cycle. All remaining nontrivial corresponding components are complete ported cycle or theta factors. Physical localization proposition 7.4 fixes one target incoming/completion type for each source-open focal germ and excludes remote compensation. The complete local classification theorem 9.3 then makes each such pair either labelled-isomorphic or ordinary-triangle related. The one-/two-port transports reconstruct every segment word and make all triangle choices coherent. Thus $N \equiv_{\triangle} N'$. $\square$

It remains to realize the structural relation physically and at full local dimension.

Lemma 10.2 (Simultaneous physical bridge gluing)**.** *Suppose corresponding local factors have one common projective K3P germ, through isomorphism or lemma 5.3. After shrinking finitely many local germs, their contractions along the common bridge tree form a common full-dimensional regular global germ, both on $\mathcal{D}_{3,+}$ and on $\mathcal{D}_{3,\mathrm{CT}}$.*

Proof. Relative to fixed analytic bridge slices, let $A_h = a^h_{u,e}a^h_{v,e}$ be the endpoint incidence product on either network and any bridge, with $h = C, G, T$. On compactly contained local germs choose uniform bounds $0 < L \le A_h \le U$. Put

$$\varepsilon = \min\left\{\frac{1}{4}, \frac{L^2}{8U}\right\}, \qquad z_C = z_G = z_T = \varepsilon, \qquad x_h = \frac{\varepsilon}{A_h}.$$

Here z is the effective bridge coordinate and x is the actual physical bridge spectrum. Since $0 < \varepsilon \le 1/4$, the effective isotropic coordinate lies in $\mathcal{D}_{3,\mathrm{CT}}$: its principal margins are at least $1-\varepsilon \ge 3/4$, and its continuous-time margins are at least $\varepsilon - \varepsilon^2 \ge 3\varepsilon/4$. Moreover

$$\frac{\varepsilon}{U} \le x_h \le \frac{\varepsilon}{L} \le \frac{1}{8}.$$

Indeed, $\varepsilon \le L^2/(8U)$ and $L \le U$, so $\varepsilon/L \le L/(8U) \le 1/8$. Every principal-domain composition margin of x is therefore at least $1-2\varepsilon/L \ge 3/4$. Every continuous-time margin is at least, cyclically,

$$x_C - x_G x_T \ge \frac{\varepsilon}{U} - \frac{\varepsilon^2}{L^2} \ge \frac{7\varepsilon}{8U} > 0.$$

Thus the same effective isotropic bridge spectrum works simultaneously for both networks and all bridges. Incidence factors cancel in (7). These finitely many inequalities are uniform and strict, so a common three-dimensional neighborhood of every z keeps both effective and actual spectra in $\mathcal{D}_{3,\mathrm{CT}}$ and supplies three independent bridge directions. The physical product extraction in lemma 6.2 is a local inverse, so the contraction has the expected rank and the common image germ is full-dimensional. $\square$

Proof of theorem 2.1: sufficiency. Assume $N \equiv_\triangle N'$. Transport physical parameters across every labelled isomorphism. For each redirected triangle use the same smooth relative H_{14} germ and its physical analytic sections; choose the generic rank once in the joint contextual contraction, as in lemma 5.3. Apply lemma 10.2 to all bridges simultaneously. The resulting common germ is regular and full-dimensional in both complete network images, so $N \bowtie_{3,+} N'$. A common germ with physical sections gives the directed relation in both directions by definition. This proves (1). Since $\equiv_\triangle$ is symmetric, proper one-sided containment is impossible. $\square$

The triangle part of this proof is full-dimensional *relative to each network image.* No step replaces H_{14} by an ambient-open 15-dimensional triangle model.

11 Genericity and reconstruction

Let

$$\mathcal{V}_N = \overline{\Phi_N(\mathbb{C}^{m_N})}^{Z}$$

be the complex Zariski closure of the polynomial K3P map. It is irreducible as the closure of the image of an irreducible affine space. Let r be the generic complex Jacobian rank. Every Jacobian minor has real-polynomial coefficients and some r-minor is nonzero. Such a polynomial cannot vanish throughout the Euclidean-open physical domain, so $d_N \geq r$; the reverse inequality is immediate. The generic-rank theorem therefore gives $\dim \mathcal{V}_N = r = d_N$.

Proof of theorem 2.2. For fixed leaf count n, only finitely many labelled networks occur in the class. Indeed, choosing tree or leaf children from the root and from each reticulation produces $r+1$ disjoint terminal tree paths, so $r \leq n-1$. If t is the number of nonroot tree vertices, degree counting gives $t = n + r - 2$, and the rooted presentation has

$$1 + t + r + n = 2n + 2r - 1 \leq 4n - 3$$

vertices.

For a competitor $N' \not\equiv_{\triangle} N$, put

$$C_{N,N'} = \mathcal{M}_{3,+}(N) \cap \mathcal{M}_{3,+}(N')$$

and form the physical target incidence correspondence

$$Z_{N'} = \{(q, \theta') : q = \Phi_{N'}(\theta'),\ q \in \mathcal{M}_{3,+}(N),\ \theta' \in \Theta_{3,+}(N')\}.$$

These sets and their projections are semialgebraic by Tarski–Seidenberg [3, Theorem 2.2.1]. Define the total source rank-drop locus

$$R_N = \{\theta \in \Theta_{3,+}(N) : \operatorname{rank} D\Phi_N(\theta) < d_N\}.$$

Choose a finite Nash stratification $R_N = \bigsqcup_\alpha S_\alpha$ on which every restricted map $\Phi_N|_{S_\alpha}$ has constant rank. Since $T_\theta S_\alpha \subseteq T_\theta \Theta_{3,+}(N)$,

$$\operatorname{rank} D(\Phi_N|_{S_\alpha})(\theta) \leq \operatorname{rank} D\Phi_N(\theta) \leq d_N - 1.$$

The constant-rank theorem and the fact that semialgebraic maps do not increase dimension give $\dim \Phi_N(R_N) \leq d_N - 1$ [3, Theorem 2.8.8]. Thus a d_N-dimensional physical intersection cannot be confined to the image of rank-deficient source parameters.

Stratify the regular source image, the target parameter space, and $Z_{N'}$ compatibly with all relevant rank loci, refining so that every restricted projection has constant rank. If $C_{N,N'}$ had dimension d_N, then lemma 7.3 would place a nonempty relatively open germ of a regular d_N-dimensional source-image stratum in that intersection. Some incidence stratum would project onto a smaller such germ with rank d_N. The analytic constant-rank theorem supplies a physical real-analytic right inverse

$$s : q \longmapsto \theta'(q), \qquad \Phi_{N'}(s(q)) = q,$$

after shrinking the output germ. Choose a regular source point above it and shrink a source-parameter neighborhood U so that $\Phi_N(U)$ lies in the section domain. Then

$$\sigma = s \circ \Phi_N : U \longrightarrow \Theta_{3,+}(N')$$

is physical and real analytic and satisfies $\Phi_N = \Phi_{N'} \circ \sigma$ on U. This is exactly $N \preceq_{3,+} N'$, contradicting theorem 2.1. Hence every competing physical intersection has semialgebraic dimension at most $d_N - 1$.

By [3, Proposition 2.8.2], a semialgebraic set and its real Zariski closure have the same dimension; finite unions take the maximum of the member dimensions [3, Proposition 2.8.5(i)]. For completeness, if $A = \mathbb{R}[x]/I_{\mathbb{R}}(C)$ is the coordinate ring of such a real closure, then complexification gives the finite faithfully flat integral extension $A \to A \otimes_{\mathbb{R}} \mathbb{C}$, which preserves Krull dimension. Therefore the complex Zariski closures of both $C_{N,N'}$ and $\Phi_N(R_N)$ inside $\mathcal{V}_N$ have dimension at most $d_N - 1$ and are proper.

Define $E_N \subset \mathcal{V}_N$ to be the following finite union (there are only finitely many binary network topologies on the fixed labelled leaf set):

(i) the complex Zariski closure in $\mathcal{V}_N$ of $\Phi_N(R_N)$;
(ii) the complex Zariski closures in $\mathcal{V}_N$ of $C_{N,N'}$, over all competitors $N' \not\equiv_\triangle N$;
(iii) the singular locus of $\mathcal{V}_N$; and
(iv) the finitely many reconstruction-test hypersurfaces $\mathcal{V}_N \cap Z(P)$ used by R1–R8 for which $P \circ \Phi_N$ is certified nonidentically zero on the intended source stratum.

Each member is proper. A finite union of proper closed sets cannot fill the irreducible variety $\mathcal{V}_N$, so E_N is proper. Outside it, every realizing topology is triangle-equivalent to N. □

The generic cut step in this proof uses proposition 4.5, not a universal pointwise converse. Since there are finitely many candidate splits for fixed N, all noncut minors are simultaneously nonzero outside a proper algebraic subset.

Proof of theorem 2.3. Given an exact Fourier tensor outside E_N, apply the following finite procedure.

R1. Fourier-transform the exact pattern tensor.
R2. Test the finitely many split flattenings. By propositions 4.3 and 4.5, recover the compatible bridge-split system and build the labelled abstract bridge-incidence tree.
R3. Use the six-circuit sum of squares to distinguish ordinary degree-three components from ordinary sunlets. This test precedes bridge normalization because every circuit is a semi-invariant under the positive incidence gauges: both monomials acquire the same nonzero factor, so its zero/nonzero value is already defined on the projective local tensor orbit.
R4. Factor positive bridge blocks in the C, G, T sectors and impose (10); recover each projective local tensor orbit and normalized effective bridge spectrum.
R5. Enumerate the finitely many rigid-support candidates compatible with the recovered tree and apply the exact pointwise, quartic, and rank tests.
R6. Follow only the exact fixed-full restoration records and their stored transports.
R7. Use the one-port segment deck and two-port order deck to recover every labelled subdivision word. No selected tensor uses more than nine physical ports.
R8. Assemble coherent mixed-graph candidates and group them into ordinary triangle classes. For each finite class $\mathcal{C}$, decide the existential semialgebraic statement

$$q \in \bigcup_{H \in \mathcal{C}} \mathcal{M}_{3,+}(H).$$

Exactly one class is feasible outside E_N; return its canonical representative.

Every loop is finite, and real-closed-field quantifier elimination terminates under the exact-input convention. The output is a topology class, not a numerically stable estimator and not a recovery of individual edge parameters inside the bridge fibre. □

12 Strict continuous time

The strict continuous-time cone $\mathcal{D}_{3,\mathrm{CT}}$ is a nonempty Euclidean-open subset of $\mathcal{D}_{3,+}$. To see the latter inclusion directly, put

$$u = \sqrt{gt/c}, \qquad v = \sqrt{ct/g}, \qquad w = \sqrt{cg/t}.$$

Then $u, v, w \in (0, 1)$, $(c, g, t) = (vw, uw, uv)$, and

$$1 + c - g - t = 1 + vw - u(v + w) > (1 - v)(1 - w) > 0,$$

with the other two inequalities cyclic. Every generic maximal Jacobian minor has the same rank on $\mathcal{D}_{3,\mathrm{CT}}$ as on $\mathcal{D}_{3,+}$, because a nonzero polynomial cannot vanish on the nonempty open continuous-time parameter domain.

Proof of corollary 2.4. Because $\mathcal{D}_{3,\mathrm{CT}}$ is Euclidean-open in $\mathcal{D}_{3,+}$, a source-open strict continuous-time containment witness is also a principal-domain containment witness. Necessity therefore follows from theorem 2.1; the discussion below verifies that the constructive and generic ingredients also remain inside the strict continuous-time domain.

Root movement, true-cut rank, the 204 pointwise obstructions, and the tree–sunlet sum of squares hold throughout $\mathcal{D}_{3,+}$, hence throughout $\mathcal{D}_{3,\mathrm{CT}}$. The isotropic edges in the displayed-tree specialization used for generic noncut recovery are strict continuous-time points, and the selected inheritance endpoints are moved into the physical interior by continuity. Triple-product marginal sections and physical local products are supplied by the continuous-time part of lemmas 4.1 and 7.1. Each polynomial or rank witness remains nonzero on a dense open subset of $\mathcal{D}_{3,\mathrm{CT}}$.

The common triangle preimage in theorem 5.2 is strict continuous-time. The simultaneous bridge construction in lemma 10.2 has the uniform positive actual-edge lower margin $7\varepsilon/(8U)$, while the effective isotropic bridge has margin at least $3\varepsilon/4$. Thus necessity and sufficiency both restrict to $\mathcal{D}_{3,\mathrm{CT}}$, and the same semialgebraic genericity and exact reconstruction arguments apply. □

No boundary case $c = gt$, $g = ct$, $t = cg$, a zero eigenvalue, or an inheritance probability in $\{0, 1\}$ is included.

13 Sharpness

The sharpness construction is independent of the fourteen-orbit atlas. Its three-leaf base pair has rooted presentations

$$\begin{aligned}
W:&\quad rS, rL_0, SU, SV, UX, VZ, ZX, UV, ZL_1, XL_2, &&\mathrm{Ret}(W) = \{V, X\},\\
W':&\quad rS, rL_0, SU, SX_0, VX_0, UX_1, VX_1, UV, X_0L_1, X_1L_2, &&\mathrm{Ret}(W') = \{X_0, X_1\}.
\end{aligned}$$

Suppress r under definition 3.2 and retain every reticulation arrowhead.

Proposition 13.1 (Topology of the base pair)**.** *The mixed graphs W, W' are simple, binary, level two, and lie in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$. Their exhaustive rooting censuses*

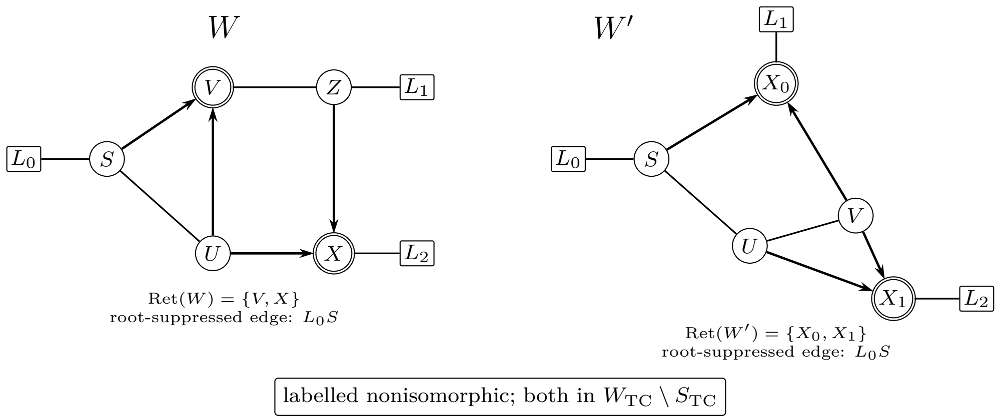


Figure 4: The sharpness base pair after the prescribed one-step suppression of the root. Plain lines are ordinary edges and arrowheads are precisely the retained reticulation entries. The drawings make the distinct labelled mixed-graph structures visible; both graphs are weakly but not strongly tree-child.

$$(\#\text{admissible}, \#\text{tree-child}, \#\text{non-tree-child})$$

are

$$W : (5, 2, 3), \qquad W' : (7, 2, 5).$$

They are labelled nonisomorphic and not ordinary-triangle equivalent.

Figure 4 makes the distinct underlying labelled graphs and the weak-but-not-strong obstruction visible before the exact rooting census is invoked.

Proof. Exact enumeration orients every ordinary edge for every candidate root edge, then checks binary degree, acyclicity, reachability, the lowest-stable-ancestor condition by dominators, and tree-childness. The displayed rooting supplies a tree-child witness and another admissible rooting leaves an omnian, giving weak but not strong status. Exact labelled mixed-graph comparison finds no isomorphism. Even after all internal arrowhead flags are forgotten, the underlying labelled graphs have no isomorphism; triangle redirection cannot change the underlying graph. □

13.1 The certified common rank-15 point

Let $y \in \mathbb{R}^{15}$ be the scaled pivot variables in the equality system

$$F_i(y) = s_i\big(q_i(W; x_W(y)) - q_i(W'; x_{W'}(y))\big) = 0, \qquad 1 \le i \le 15,$$

for the fifteen nonconstant conservation-supported three-leaf K3P coordinates. The literal maps have 64 direct parameters: ten edge spectra and two inheritance probabilities on each network. Fifteen are pivots and the other 49 are frozen at exact rational values.

Proposition 13.2 (Krawczyk sharpness certificate)**.** *There is a rational center y_0 and the box*

$$X = \prod_{j=1}^{15} [y_{0,j} - 10^{-50}, y_{0,j} + 10^{-50}]$$

such that:

(i) the Krawczyk operator maps X strictly into its interior and the equality system has a unique root in X;

(ii) both normalized network maps have rank 15 throughout X;

(iii) all edge spectra, transition probabilities, inheritance probabilities, and strict continuous-time margins remain positive throughout X.

At the common tensor, the two images share an ambient-open regular 15-dimensional germ.

Proof. Exact rational Gauss–Jordan elimination constructs $Y = J_F(y_0)^{-1}$. Directed rational interval arithmetic encloses $J_F(X)$ and forms

$$K(y_0, X) = y_0 - YF(y_0) + (I - YJ_F(X))(X - y_0).$$

Every component lies strictly inside X. The largest normalized distance from the box center is $9.74099938 \times 10^{-41}$, and

$$\|I - YJ_F(X)\|_\infty \leq 8.07702308 \times 10^{-47} < 1.$$

The inclusion gives existence; the norm bound makes every mean Jacobian in the convex box invertible and gives uniqueness in this equality slice.

For each 15-by-32 local network Jacobian, exact row reduction selects a 15-minor. Neumann bounds throughout X are at most $1.54315210 \times 10^{-45}$ for W and $4.58271952 \times 10^{-45}$ for W', excluding zero. The smallest exact continuous-time lower bounds are positive (approximately 4.96×10^{-10} and 1.40×10^{-9}). The complete certificate stores rational intervals for every inequality, not only these summaries. The submersion theorem now gives an ambient-open output neighborhood for each map, whose intersection is the common regular germ. □

The uniqueness in proposition 13.2 concerns only the selected 15-variable equality slice; it is not global numerical-parameter identifiability.

13.2 Identical K3P cherry substitution

Replace the same labelled leaf on both networks by the same labelled cherry. Let the new pendant spectra be $u = (u_C, u_G, u_T)$ and $v = (v_C, v_G, v_T)$. For each nonzero character h, choose a retained outside leaf j. If q' is the enlarged tensor and ℓ is the old leaf replaced by the cherry a, b, Fourier conservation gives the explicit observables

$$R_h = \frac{q'_{j=h,\,a=h,\,b=0,\,\text{others}=0}}{q'_{j=h,\,a=0,\,b=h,\,\text{others}=0}} = \frac{u_h q_{\ell=h,\,j=h,\,\text{others}=0}}{v_h q_{\ell=h,\,j=h,\,\text{others}=0}} = \frac{u_h}{v_h},$$
$$P_h = q'_{a=h,\,b=h,\,\text{others}=0} = u_h v_h q_{\ell=0,\,\text{others}=0} = u_h v_h.$$

The common old-network factor in the ratio is strictly positive, and the last equality uses $q_{\mathbf{0}} = 1$.

Lemma 13.3 (Six-dimensional cherry inverse)**.** *On the positive branch,*

$$\det \frac{\partial(R_C, P_C, R_G, P_G, R_T, P_T)}{\partial(u_C, v_C, u_G, v_G, u_T, v_T)} = \frac{8u_C u_G u_T}{v_C v_G v_T} \neq 0.$$

Thus one identical cherry adds exactly six dimensions to a common regular germ. At

$$u = (2/5, 4/9, 3/7), \qquad v = (3/7, 5/11, 4/9),$$

both spectra are strict continuous-time and the determinant is $176/25$*.*

Proof. The determinant factors into three 2-by-2 sector blocks. Its direct Laurent-polynomial expansion gives the displayed formula. The inverse is

$$u_h = \sqrt{R_h P_h}, \qquad v_h = \sqrt{P_h/R_h}.$$

After recovering these six coordinates, divide retained-cherry Fourier coordinates by their nonzero pendant factors to recover the old tensor. Thus the new map has at most six new parameter directions and at least six locally observable directions. □

Proof of theorem 2.5. The base case follows from propositions 13.1 and 13.2. Repeatedly apply lemma 13.3. A tree-child base rooting lifts; the old omnian and its incident arcs remain unchanged; the cherry stem (replacing the former pendant edge) and the two new pendant edges are bridges, so the blobs and level remain unchanged. Contracting the uniquely labelled newest cherry recovers the previous graph. Hence any enlarged isomorphism or triangle equivalence would descend to one for the base pair, which does not exist. The dimension is $15 + 6(n-3) = 6n - 3$. □

Therefore "strongly" in theorem 2.1 cannot be weakened to "weakly," even in strict continuous time. The theorem does not say that every weakly tree-child network is ambiguous.

14 Outer-class tree–double-theta obstruction

The separate tree–theta construction of [17] uses a three-leaf level-2 theta blob with two reticulations and four switchings. At an exact algebraic point its core mixture factors as

$$M_{y,z} = P_{y+z} B_y B_z,$$

so, after including the common pendant factors, the network tensor is a three-leaf tree tensor. All inverse-Fourier transition entries are strict. The fixed normalized theta map has a nonzero 15-by-15 Jacobian determinant

$$\frac{h(10h^2+1)}{2^{61}3^4 5^{14}} > 0, \qquad h = 5^{-1/4},$$

whereas the tree map has rank 9. The preimage theorem therefore makes the theta-parameter locus mapping into the tree model a smooth 23-dimensional submanifold of codimension six. A certified tangent and the real-analytic implicit-function theorem perturb the collision into the strict continuous-time cone.

Because the theta map is a submersion at the common point, its image contains an ambient-open neighborhood. A smaller tree output germ lies inside that neighborhood and has a physical analytic theta section, proving the proper directed containment in proposition 2.6.

This example does not contradict theorem 2.1. Exhaustive root enumeration gives the theta mixed graph census $(7, 0, 7)$: it has no tree-child admissible rooting and lies outside W_{TC}. It is therefore an outer obstruction, not the sharp transition from strong to weak tree-childness. It also explains why no proof here treats arbitrary three-leaf nontrivial blobs as pointwise detectable.

15 A unified Kimura-family perspective

The K3P theorem and the same-author JC and K2P companion manuscripts report the same structural quotient but different local analytic geometry. The latter two are unreviewed companion results, not independent corroboration. The comparison below records only the statements and exact packages of these three programs [19, 18]; it is not an extrapolation to other group-based models.

feature	JC	K2P	K3P
independent nonzero character sectors	1	2	3
bridge scales per component incidence	1	2	3
ordinary-triangle normalized local dimension	4	9	14
triangle image near the common point	ambient	ambient	quartic H_{14}
certified weak-class ambiguity dimension	$2n+1$, $n \geq 4$	$4n-3$, $n \geq 3$	$6n-3$, $n \geq 3$
strong-class topology quotient	$\equiv_\triangle$	$\equiv_\triangle$	$\equiv_\triangle$
proper directed containment in strong class	none	none	none

K3P therefore agrees topologically with the two symmetry-reduced models but not geometrically. Its ordinary triangle loses one ambient dimension even though its three Fourier sectors are independent; the common ambiguity is the smooth part of an irreducible quartic hypersurface. Conversely, each identical cherry exposes six new directions, two per sector, accounting for the all-n sharpness dimension.

Within these three same-author programs the reported conceptual boundary is strong tree-childness: inside it, positive bridge factorization, local factor classification, and coherent probes leave only ordinary triangle redirection; immediately outside it, robust full-dimensional ambiguities occur. This synthesis is restricted to JC, K2P, and K3P on the domains and network class stated in their respective theorems. No independent-consensus claim, or theorem for arbitrary finite groups, arbitrary state-label symmetries, or signed Fourier components, is asserted.

16 Scope and limitations

The result is an exact infinite-data information theorem. It assumes a site-pattern tensor generated exactly by the stated model and classifies the semi-directed topology, not all numerical parameters.

In particular, incidence gauges in theorem 6.1 prevent the topology theorem from identifying individual physical bridge multipliers.

The theorem is restricted to:

- binary standard semi-directed networks under the fixed one-step root-suppression convention;
- level at most two;
- strong tree-childness for the classification;
- positive K3P Fourier eigenvalues and strictly positive transition probabilities;
- inheritance probabilities strictly between zero and one;
- fixed observable state labels C, G, T; and
- the principal positive domain, or separately the strict continuous-time cone.

It does not cover nonbinary networks, higher level, arbitrary weakly tree-child networks, boundary probabilities, zero or signed eigenvalues, an untransported permutation of the three nonzero states, or arbitrary nonreversible substitution models. No claim is made for the signed K3P components outside $\mathcal{D}_{3,+}$.

Generic identifiability means outside a proper algebraic exceptional set for each fixed topology. It is not pointwise topology identifiability at every parameter, equality of complete stochastic images, or universal identifiability of physical parameters. The exact reconstruction theorem is a termination result under an exact-real oracle. It supplies no finite-sample estimator, statistical error rate, numerical conditioning bound, or practical sequence-length guarantee.

Strong tree-childness is sharp against weakening to weak tree-childness: the weak-class construction shows that this weakening makes the classification false. It does not classify every weakly tree-child network or every possible weak-class ambiguity.

17 Reproducibility

The proof has a finite machine dependency. Exact arithmetic is used for model-domain identities, graph transports, polynomial pullbacks, Jacobian minors, rank factorizations, restoration, and probes. Rigorous rational interval arithmetic is used for the Krawczyk sharpness theorem. The certificate package records primitive graph inputs, parameter and coordinate orders, source–target direction, physical margins, evidence hashes, and reproduction commands.

The primary verifier passes all 28 entries in its primary chain. Separately, the integrated classification report binds 85 active artifacts and verifies the equivalence in equation (1), and its fail-closed suite rejects all 27 integrated mutations. These binding checks are supplemented by the component-specific clean-room, interval, restoration, probe, cut-transfer, and global-infrastructure replays described in the reader supplement.

The primary implementation and separately structured replays do not accept a stored final Boolean. Their exact boundaries are nevertheless different and are stated rather than collapsed into one notion of independence. The K3P algebraic replays rebuild Fourier tensors, coordinate transports, polynomial pullbacks, rank witnesses, and physical inequalities. The clean-room H21 audit preserves and reproduces the historical rooted-DAG automorphism failure, then validates the corrected root-suppressed mixed-graph transport; its hardened 25-mutation re-audit is an active command. A complete four-port producer visits all 405,216 primitive directed presentations and derives the 27,834 post-topology presentations and the final $40 = 38+2$ residue. A separately implemented verifier imports neither that producer nor its graph atlas: it reconstructs the primitive grammar, exact maps, topology tests, rank minors and upper bounds, polynomial pullbacks, restoration bindings, and the fourteen-orbit quotient. For cut transfer, an active standalone graph producer independently derives the five primitive templates, 72 labelled records, and literal switching signatures; its fresh output must match byte for byte the input consumed by the K3P algebraic replay. From that graph-derived table the separately implemented K3P verifier recomputes displayed flags, all 204 normalized directions, and every exact obstruction. A separate model-independent enumeration visits all 808,642 balanced noncut boundary words and a no-import reduced-palette replay checks 379,742 valid presentations with zero all-switching survivors. The displayed tree witness then supplies an explicit nonzero 5×5 minor, so no companion JC cut theorem is used. Coherent graph-subdivision and mask-rebinding mutations test both the semantic reconstruction and the byte lock. The global transfer audit checks that no common bridge tree or fourteen-orbit result is assumed. The sharpness replay derives a new exact preconditioner and verifies every interval inequality throughout the box.

The hour-scale probe producer enumerates 574,535 one- and two-port rows. Its separate full semantic replay reconstructs the public anchor graphs and all insertions, restrictions, transports, quartet splits, row-specific tree–sunlet circuit decks, reverse marginals, and triangle inheritance. It imports neither the probe producer nor the historical atlas. The current registry declares no Bernstein sign certificates; the replay nevertheless implements the generic exact Bernstein test and a mixed-sign mutation requires rejection rather than vacuous acceptance.

Before those probe rows are generated, a separate graph-only gate derives the complete 133-record tree/cycle/θ_2 starting universe from the literal grammar and fixed-full restorations. Its no-import verifier independently reconstructs that census, and a crosswalk combines it with the 43 anchors from the frozen canonical four-port serialization, each independently checked against the exhaustive full four-port replay, and requires exact agreement with all 176 public contract rows. It also reconciles all 424 physical descendants of the 176 marginalized-incoming theta presentations with 66 existing isomorphic one-port rows, with zero unmatched paths. Omission, duplication, relation change, graph-hash drift, incoming-boundary drift, and folded-transport mutations fail closed.

Targeted mutation suites reject, among other changes, deletion or duplication of relation rows, source–target reversal, changed ports or incoming roles, quartic reassignment, an omitted H_{14} monomial, an untransported sector permutation, the false equality $C = T$, weakened rank bounds, missing restoration children, broken probe parentage, inconsistent triangle choices, boundary inheritance probabilities, interval determinants containing zero, and Krawczyk boxes that lose strict self-mapping. Optimized interpreter mode is refused at verifiers whose safety must not depend on assertions.

Historical, superseded, or exploratory artifacts are excluded from the active evidence manifest. In particular, the earlier universal pointwise K3P cut-recovery claim is withdrawn and not used; the active proof uses the three statements separated in section 4. “Independent replay” means a separately implemented program, not review by an independent human specialist.

Data and code availability

The exact certificate and replay snapshot used for this revision is available in the public repository at immutable commit `203e114ace0ead3852f109a3713acda37bf74e65`, under `k3p_level2_identifiability_final`. The reader supplement gives a relative-path theorem-to-certificate crosswalk. This manuscript does not assert a repository release DOI or a paper/data/code license; neither has been supplied for this workstream.

Author contributions

Alec Kriebel conceived and directed the study, fixed its mathematical scope and conventions, curated the exact reproducibility package, adjudicated the proof and audit results, and accepts responsibility for the claims, software, and submission.

Use of generative AI

Generative AI systems assisted with mathematical exploration, code generation, implementation, drafting, and adversarial review. Automated claims were not accepted as evidence: active claims are bound to exact graph-derived certificates, direct derivations, separately implemented replays, rigorous interval arithmetic where used, and fail-closed mutation tests. No independent human specialist review is claimed. The author accepts responsibility for the mathematical statement, code, and final submission.

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